

Data visualization guide

How McKesson Workforce Innovation turns workforce data into charts that lead to decisions.

Version 0.2 (pilot) September 2026 All example data is synthetic

OVERVIEW

How to use this guide

One set of rules, three ways in. The guide teaches the thinking; the starter kit makes the right answer the easy one in every tool we use.

Every chart we make should do one job: answer a business question so clearly that the reader knows what to do next. The rules here are written to be checked, by a reviewer or by an agent, before a chart leaves the team.

Inspired by the structure of a leading consulting firm's public data-visualization guide, rebuilt for McKesson's brand and for the work of Workforce Innovation.

CONSULTANTS AND ANALYSTS

Decks, reports, workbooks

Read Fundamentals once, then use the chart pages as you build. Keep the cheat sheet next to you.

- PowerPoint template and starter deck
- Excel chart workbook with the palette
- One-page cheat sheet and chart chooser
- Power BI report theme

DEVELOPERS AND DATA SCIENTISTS

Python, HTML, R, Power BI

Each chart page links to tested code. Themes are generated from one token file, so every tool matches.

- `mckviz` for matplotlib and Plotly
- CSS variables, Plotly and Vega-Lite themes
- `theme_mck()` for ggplot2
- `mv.lint(fig)` to check a chart

CODING AGENTS

Claude, Copilot and others

Agents read `AGENTS.md` automatically in this repo. For other repos, copy one self-contained rules file.

- `AGENTS.md`: workflow and rules
- `mck-chart` skill: question to QA
- `dist/agents/mck-viz-rules.md`
- Cookbook pages with runnable snippets

The five steps behind every chart

1 Question. Write the question the chart answers in one sentence.

2 Chart. Pick the chart from the question type, not from habit.

3 Build. Grey first, then color only what the headline names.

4 Headline. Compute the finding, then write it as a sentence.

5 QA. Run the checklist and look at the rendered chart.

OVERVIEW

Contents

Fundamentals apply to every chart. Each chart family then gets the same treatment: anatomy, variants, best practice and how to build it.

This is the pilot release. Sections marked NEXT follow the same pattern once the look and depth of the pilot are agreed.

01 **Fundamentals**
Question first, how readers read, anatomy, headlines, color, type, clutter, honest scales, people data, accessibility

02 **Magnitude**
Bar and column charts: anatomy, variants, best practice, build it

03 **Part-to-whole**
Stacked bars, waterfalls, treemaps, donuts: anatomy, variants, best practice, build it

04 **Change over time**
Line, stacked area, bubble timeline, small multiples

05 Ranking NEXT
Slope, bump, dot plot and dumbbell, heat table

06 Distribution NEXT
Histogram, strip, box plot

07 Correlation NEXT
Scatter and bubble, heatmaps, why not dual axes

08 Relationships and flows NEXT
Sankey, parallel coordinates, Venn

09 Geo-spatial NEXT
Choropleth, bubble map, dot density

10 Single number NEXT
KPI and big number, progress to target

11 Formats NEXT
Slides, reports and print, web and mobile, dashboards, interactivity, animation, social

12 Toolkit NEXT
Downloads, set-up for each tool, the QA checklist, guidance for agents

Start with the question, not the chart

A chart is finished when a reader can say what it means and what to do about it, without you in the room.

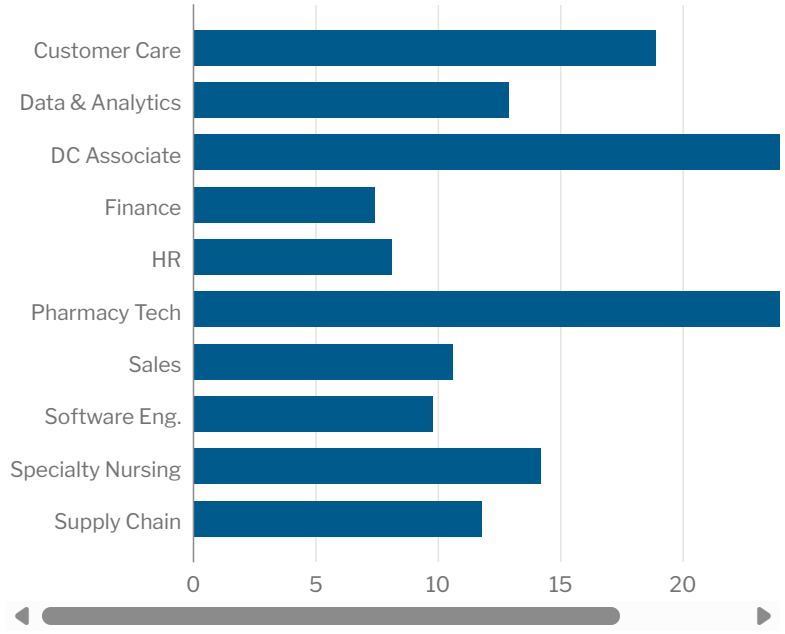
Before you open a tool, write the question in one sentence: "Where is attrition concentrated?", not "show attrition".

Then pick the mode. **Explanatory** charts prove a finding, for decks and reports (the default). **Exploratory** views let readers find answers, for dashboards. Both follow the same rules.

Both charts here are accurate; only one is useful in a meeting.

Attrition by job family

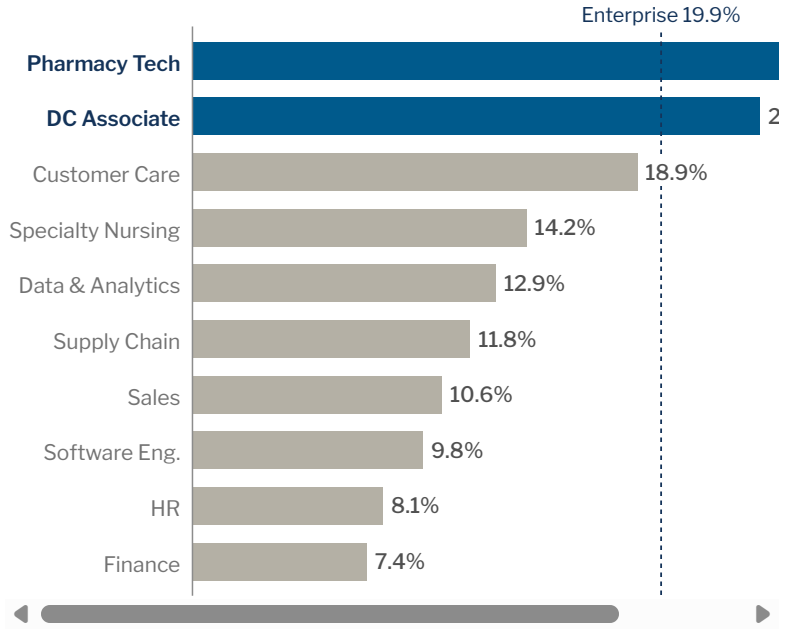
FY26



Shown
A label for a title, alphabetical order, every bar equal. The reader has to find the point alone.

Two frontline job families lose about 1 in 4 people a year, well above the 19.9% enterprise rate

Voluntary attrition by job family, FY26 (%)



Explained
The headline states the finding, the order ranks the families, and color points to the two the headline names.

How readers read a chart

Readers move through a chart in three steps. Design for each one, in order.

1 Perceive

In the first second the eye takes in shape, length and color before reading a word. Make the thing that matters the thing that stands out: one color against grey.

2 Interpret

Next the reader looks for meaning: headline, labels, units, benchmarks. The headline should confirm what they just saw, or sharpen it.

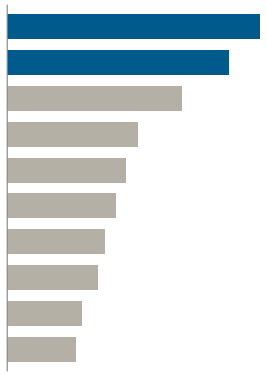
3 Act

Finally they ask "so what?". A callout that sizes the implication turns a finding into a decision.

STEP 1 · PERCEIVE

Two bars stand out

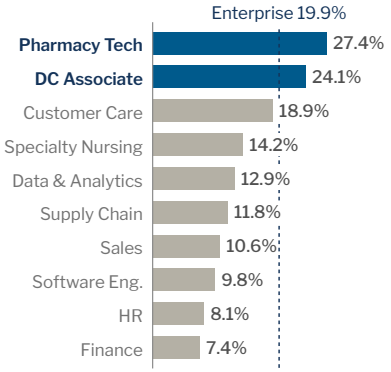
No words yet. Grey context and blue focus already tell the reader where to look.



STEP 2 · INTERPRET

Labels give it meaning

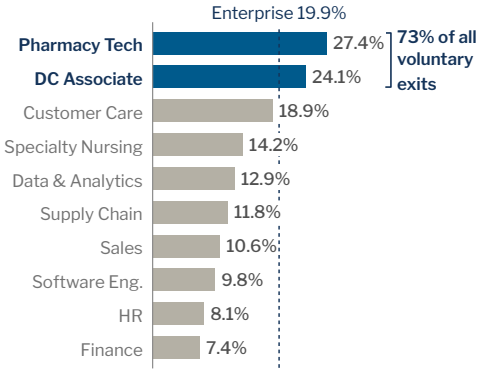
Names, values and a labelled benchmark show the two families sit well above 19.9%.



STEP 3 · ACT

The callout sizes the prize

Together they hold 58% of headcount but 73% of voluntary exits: a retention plan for two families moves the enterprise number.



Anatomy of a chart

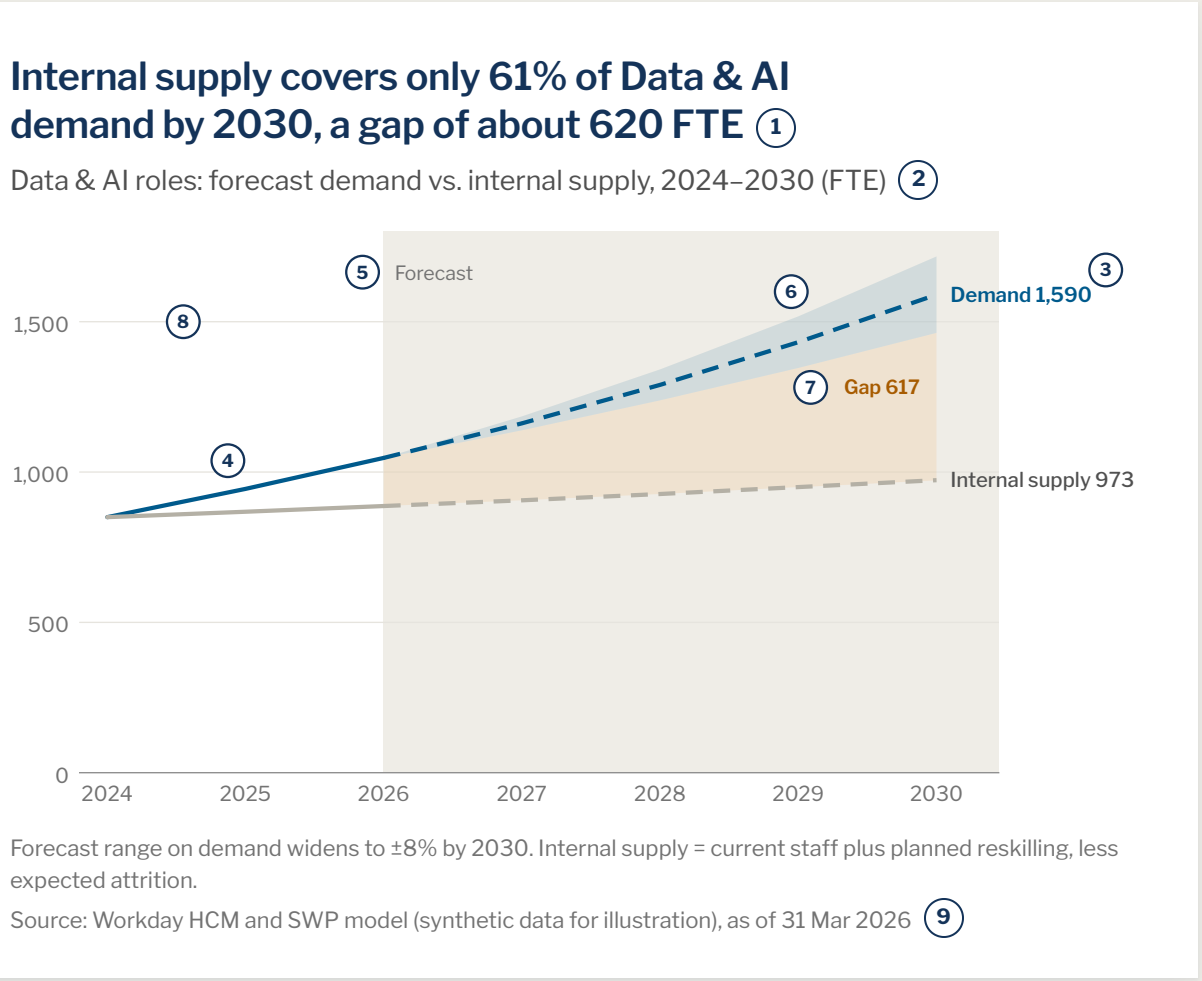
Every McKesson chart has the same four layers, top to bottom: headline, subtitle, chart, and footnote with source.

The headline and subtitle do different jobs. The headline is editorial: it states the finding. The subtitle is neutral: metric, unit, scope and period.

Inside the chart, everything is labelled in place, so the reader never has to match colors to a legend.

All four layers share one left edge. In HTML they are page text; the chart library draws only the data.

1. **Insight headline**
   A full sentence with the number. At most two lines, no final period.
2. **Subtitle**
   Metric, unit, scope and period. No opinions.
3. **Direct labels**
   Series named at the line end, in the series color. No legend.
4. **Focus and context**
   The series the headline is about is blue; the comparison is grey.
5. **Forecast marked**
   A shaded period and dashed lines separate estimates from actuals.
6. **Range, not a point**
   Forecasts show their uncertainty.
7. **The point, annotated**
   The gap the headline quotes, labelled where it is.
8. **Light scaffolding**
   Few, light gridlines; units live in the subtitle.
9. **Footnote and source**
   Definitions, system, as-of date, population.



FUNDAMENTALS

Write the headline last, from the numbers

The headline is the most-read text on the page. It should be the finding, not the topic.

The formula

Who or what + verb + number + so what.

Before you keep it

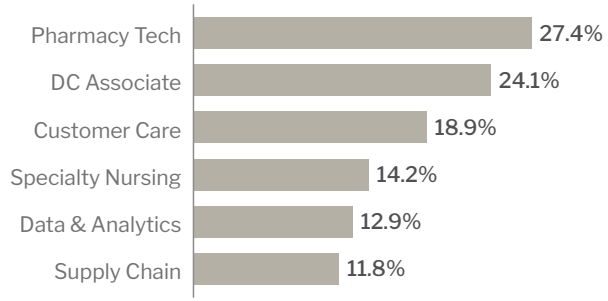
- [ ] Recompute every number, "all", "most" and "n of m" from the data
- [ ] The blue in the chart is what the headline names
- [ ] Two lines at most; no final period

If you can't write a headline, you don't have a finding yet: you have a chart.

LABEL	Attrition by job family · names the topic; says nothing
VAGUE	Attrition varies across job families · true of every chart ever made
OPINION	Attrition is out of control in the frontline · a claim the chart can't support
INSIGHT	**Two frontline job families lose about 1 in 4 people a year, well above the 19.9% enterprise rate**
INSIGHT + ACTION	**Two frontline families account for 73% of voluntary exits; retention there moves the enterprise rate**

Attrition varies across job families

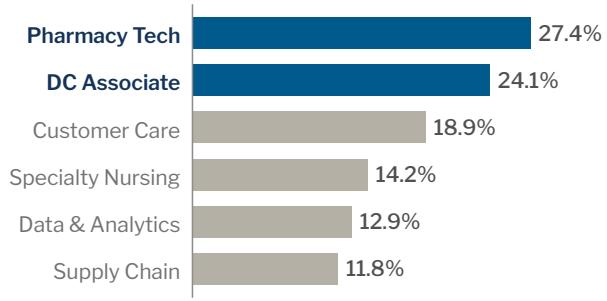
Voluntary attrition by job family, FY26 (%)



Headline and chart both neutral
Nothing tells the reader where to look.

Pharmacy and DC teams lose about 1 in 4 people a year

Voluntary attrition by job family, FY26 (%)



Headline and highlight agree
The two blue bars are exactly what the sentence names.

FUNDAMENTALS

Grey first, then color the point

Color is for meaning, never decoration. Start with every mark in grey, then color only what the headline names.

Highlight at most two items. When everything is highlighted, nothing is.

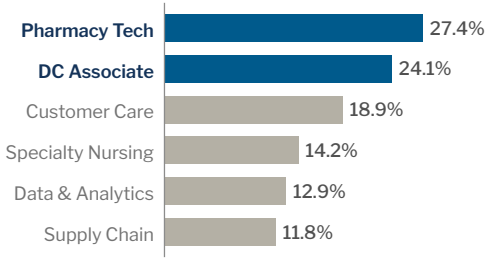
Echo the highlight in the text: bold the matching category label, and set callouts in the color of the mark they describe.

Orange means "worse". Keep it for negatives: below target, shrinking, a shortfall. Never use it as a plain highlight; readers take it as a warning.

PATTERN 1 · DEFAULT

Focus

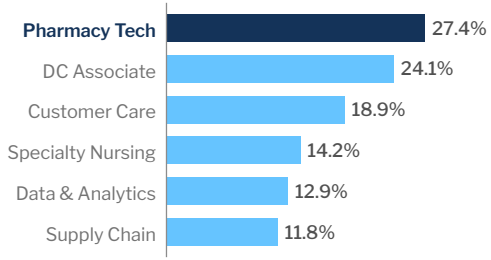
All grey; the items the headline names in blue.



PATTERN 2

Uniform + accent

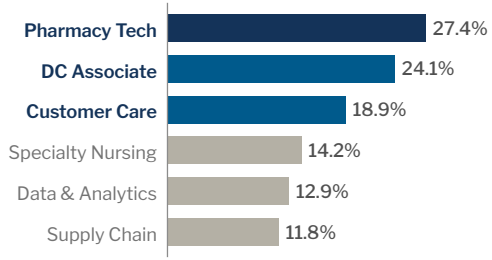
One series in light blue; the single exception in navy.



PATTERN 3

Focus + accent

Grey context, a blue group, one navy exception. Use sparingly.



Color roles

Context · everything the reader should see but not focus on	`#B4B0A5`
Focus · what the headline names	`#005A8C`
Accent · the one exception that must pop (navy, not orange)	`#143359`
Light blue · the series when navy marks the exception	`#66C4FF`
Orange · only when it means worse: below target, shrinking, a shortfall	`#EF8200`
Total and benchmark · totals, targets, reference lines	`#143359`
Context light · de-emphasized fills in stacks and areas	`#DEDBD1`

Three palettes, three jobs

Pick the palette from what the color encodes.

Categories

Only when categories must be told apart. At most six, always in the same order, so "Full-time" is the same blue on every slide.

One measure, light to dark

For heat tables and choropleths of a single measure. Darker means more.

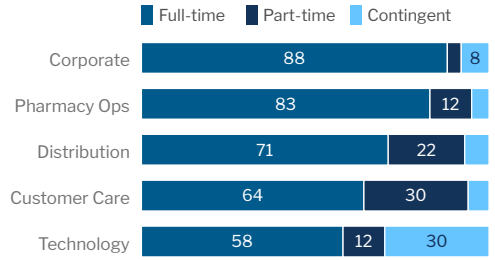
Gap to a benchmark

For values centered on zero or a target. Orange below, blue above, a neutral middle.

Text on a fill switches between white and navy, whichever contrasts more (`mv.text_on`).

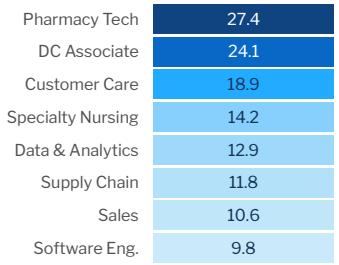
CATEGORICAL

Workforce mix by function (%)



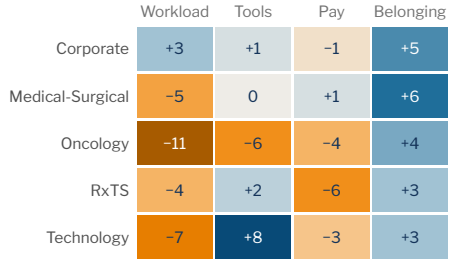
SEQUENTIAL

Voluntary attrition, FY26 (%)



DIVERGING

Engagement vs. healthcare benchmark (pts)



Categorical
Max 6, always this order

005A8C 143359 66C4FF 00AF9D 00737A EF8200

Sequential
One measure, light to dark

E6F3FB B6E2FA 66C4FF 009EFF 0A5FBC 143359

Diverging
Centered on 0 or a target

A75B00 EF8200 F7C180 EFEDE7 99BDD1 005A8C 143359

Type and proportion

One typeface, Libre Franklin, in a few fixed roles. Type should frame the data, not compete with it.

The headline is the largest text on the chart but the data is the largest *thing*. If the headline is the first thing you notice from across the room, it is too heavy.

- Nothing below 9 pt on slides, 11 px on screens
- Sentence case everywhere; no all-caps headlines
- Headline, subtitle, key, footnote and source share one left edge
- Big numbers use the light weight

Libre Franklin is open source (SIL OFL). Set-up for Office, the web and Power BI is in the Toolkit.

Role	Sample	Weight	Slide	Screen
Headline	**Attrition is concentrated in two job families**	Bold	24 pt	22 px
Subtitle	Voluntary attrition, FY26 (%)	Regular	14 pt	15 px
Data labels	27.4%	Medium	12 pt	13 px
Axis and category labels	Pharmacy Technician	Regular	11 pt	12 px
Callouts	Hiring freeze, Q2	Regular	12 pt	13 px
Footnote and source	Source: Workday HCM, as of 31 Mar 2026	Light	9 pt	11 px
Big number	75%	Light	54 pt	56 px

Proportion on a 16:9 slide

Margins of about 0.5 in. The chart takes at least 60% of the slide height. One message per chart, at most two charts per slide; the slide title is the headline.

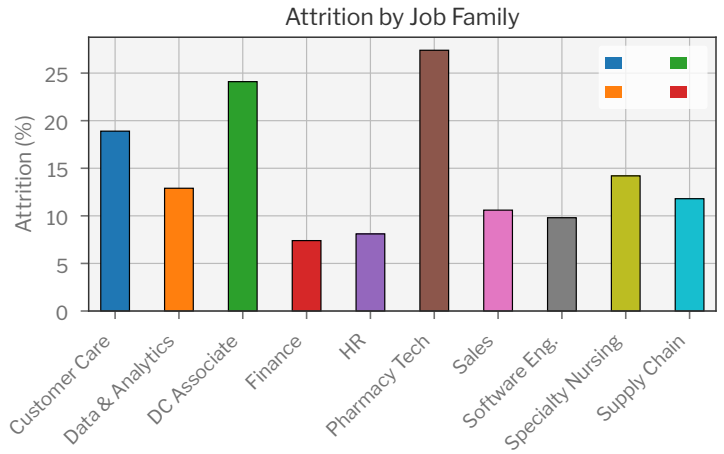
Proportion on a screen

The chart card sits on a quiet background with generous padding. Headline about 1.4× the subtitle. On phones, swap end labels for a top key and let wide exhibits scroll sideways rather than shrink.

Label directly, delete the rest

Every element must earn its place. If removing it loses nothing, remove it.

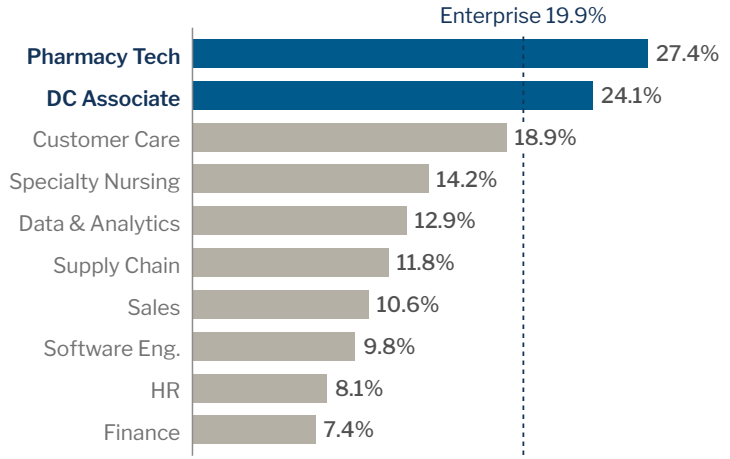
1. Replace the label-style title with an insight headline
2. Sort by value, not alphabet
3. Turn long-named columns into horizontal bars
4. Grey everything, then color the point
5. Label values in place; delete the value axis, gridlines and tick marks
6. Delete the legend, borders and background fill
7. Add the benchmark, labelled where it sits



Tool defaults
Rainbow fills, borders, rotated labels, gridlines, a legend that decodes nothing.

Two frontline job families lose about 1 in 4 people a year, well above the 19.9% enterprise rate

Voluntary attrition by job family, FY26 (%)



Decluttered
The same data. Fewer elements, and every one of them carries information.

FUNDAMENTALS

Honest scales

A chart can be accurate in every number and still mislead. These four rules prevent the common ways.

Bars start at zero. If the differences only show on a narrow range, change the chart, not the axis.

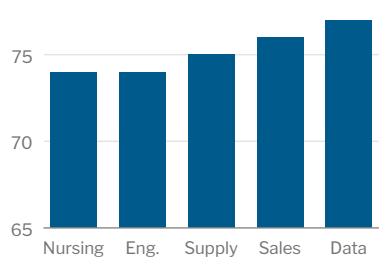
No dual axes. Two scales let you make any two lines cross wherever you like.

Bubble area, not diameter, is proportional to value.

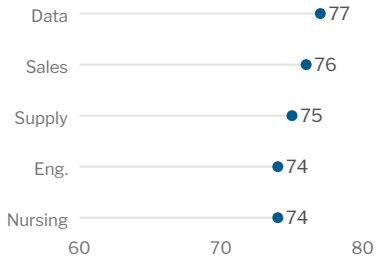
Pies take five slices at most, and sum to 100%.

The one exception: a waterfall bridging two large totals may break its axis, if the break is drawn and footnoted.

Truncated bars

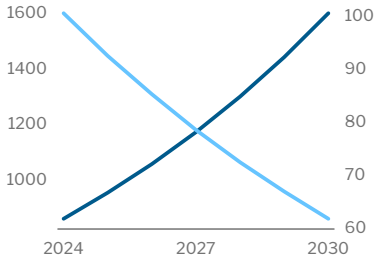


Try to avoid
An axis from 65 turns a 4% difference into a 33% one.

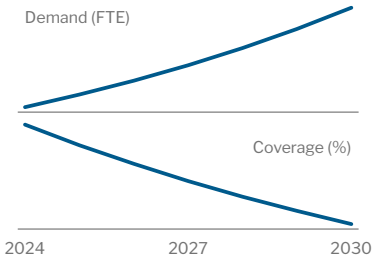


Recommended
A dot plot compares position, so the range can be narrow.

Dual axes

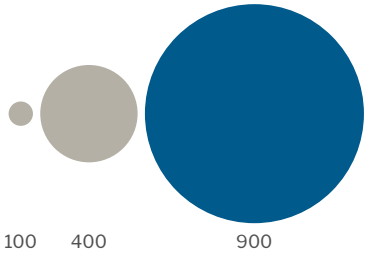


Try to avoid
The crossing point is an artifact of two chosen scales.

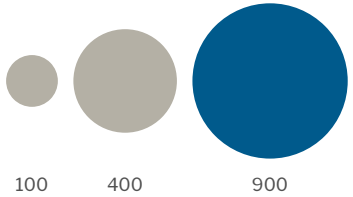


Recommended
Two aligned charts share the time axis.

Bubble size

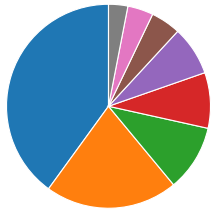


Try to avoid
Sized by diameter, 900 looks 81× bigger than 100.

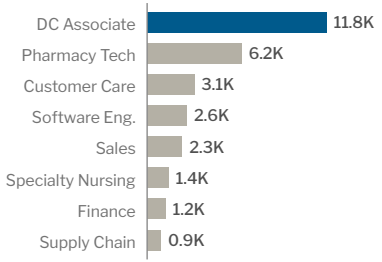


Recommended
Sized by area, 900 is 9× bigger, as it should be.

Too many slices



Try to avoid
Eight similar angles can't be ranked by eye.



Recommended
A sorted bar ranks them instantly.

People data is personal

Workforce charts describe colleagues, and carry extra obligations.

Minimum group size

No group under **10 people**: not in bars, labels, tooltips, filters or drill-downs. Merge into "Other" and footnote it.

No re-identification

Small team × tenure × location can single out one person, even when each cut is large.

Protected attributes and forecasts

Protected attributes only for approved analyses such as pay equity, aggregated. Label estimates and show a range. Frame findings about roles, not individuals.

Engagement by team, one distribution center

Two teams have fewer than 10 respondents: anyone who knows the team can guess who answered what.

Team	Respondents	Engagement
Inbound	46	71
Outbound	38	68
Returns	7	52
Quality	4	88
Site leadership	9	81

Try to avoid
Reporting groups below the minimum, even in a table or tooltip.

Engagement by team, one distribution center

Small teams combined into one group that clears the threshold.

Team	Respondents	Engagement
Inbound	46	71
Outbound	38	68
Other teams (3 combined)	20	72

Recommended
Merge into "Other" and footnote it: "Teams with fewer than 10 respondents are combined."

In dashboards, enforce it in the data model

A rule in a slide is easy to forget; a rule in the measure can't be skipped. In Power BI: `IF([Headcount] < 10, BLANK(), [Engagement])`. In Python, drop or merge small groups before plotting, and keep individual values out of hover text.

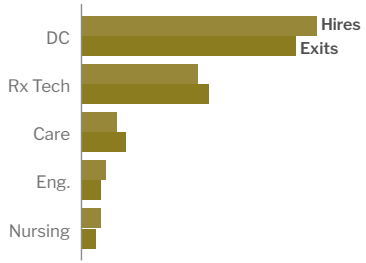
Accessible by default

About 1 in 12 men and 1 in 200 women have a color-vision deficiency. In a room of 25, assume someone does.

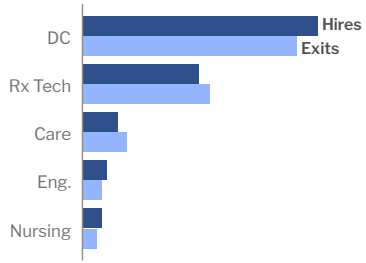
- Text contrast at least 4.5:1
- Never red vs. green; use blues and navy, with orange only for "worse"
- Never color alone: pair it with labels, position or shape
- Nothing below 9 pt on slides or 11 px on screens
- Alt text for every chart that leaves the team: the headline plus two or three key numbers

The red–green examples are simulated for deuteranopia, the most common form.

Red vs. green

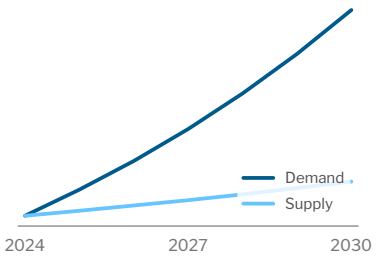


As seen with deuteranopia
Red and green hires vs. exits become one color.

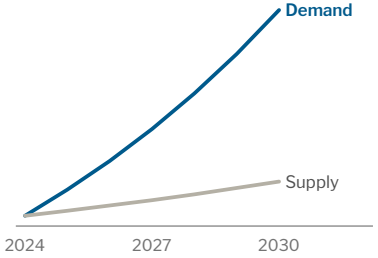


As seen with deuteranopia
Blue and light blue stay distinct.

Color as the only cue

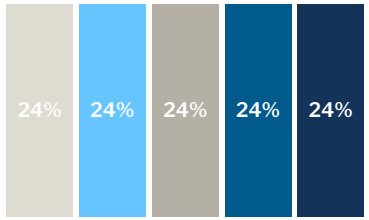


Try to avoid
The reader matches colors to a legend.

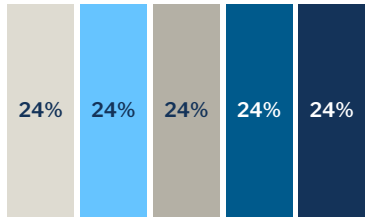


Recommended
Names at the line ends; color only reinforces.

Text on fills



Try to avoid
White on every fill fails on the light ones.



Recommended
White or navy, whichever contrasts more.

Alt text

Write it from the headline and the numbers that prove it.

> **Alt:** Two frontline job families lose about 1 in 4 people a year, well above the 19.9% enterprise rate. Pharmacy Technician 27.4%; Distribution Center Associate 24.1%; all other job families are below the enterprise rate.

In PowerPoint: right-click the chart, View Alt Text. In HTML: `aria-label` on the figure.

MAGNITUDE

Bar chart

Magnitude charts compare size: which is bigger (relative), and by how much (absolute).

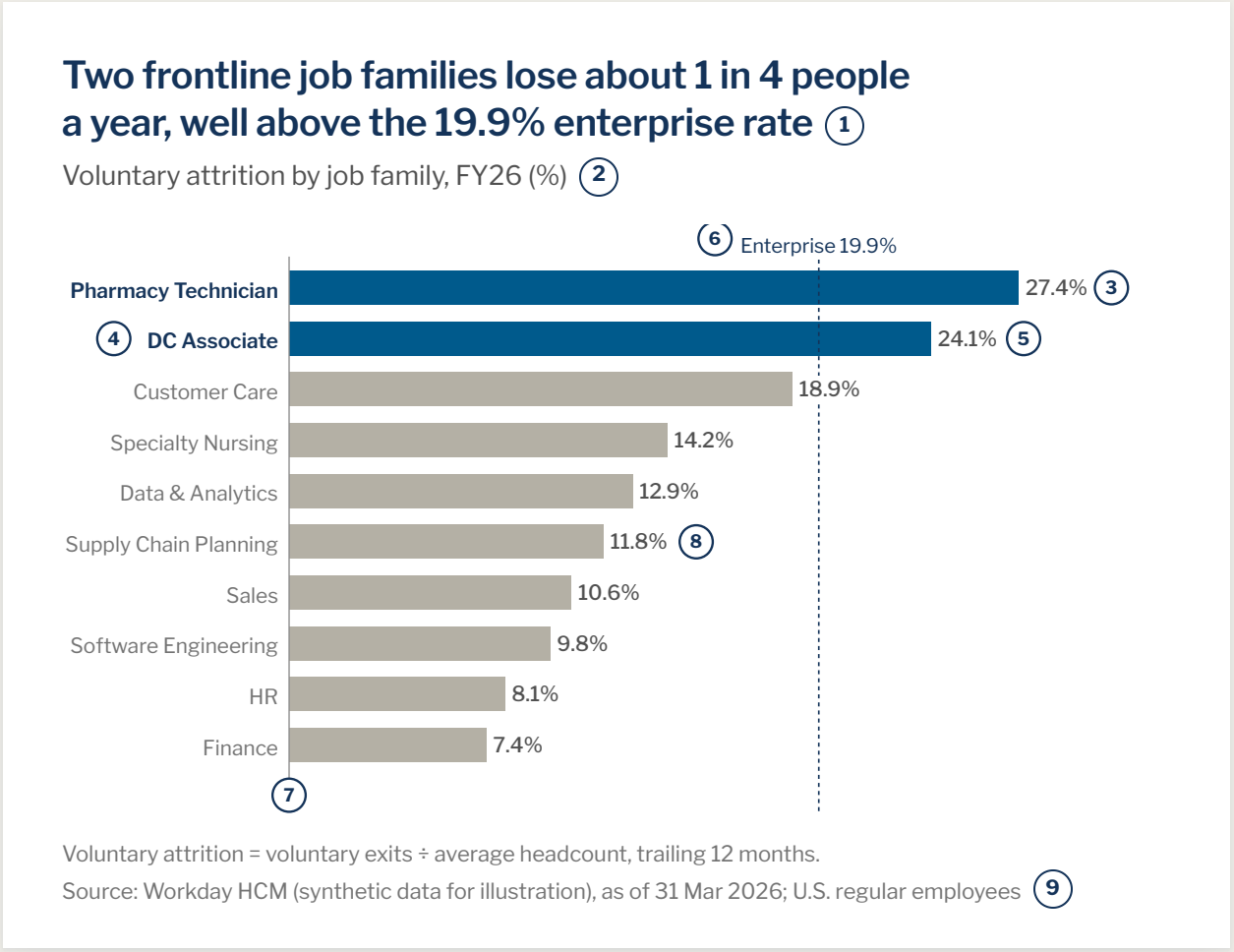
A bar's length is proportional to its value, so bars are the most precise way to compare categories. Sorted horizontal bars are the default for ranking job families, sites or skills: long names stay readable, and the layout works on a phone.

Use columns for a few time periods or short names.

McKesson spec

- Bar about 2× the gap (Excel gap width 50%)
- Sorted by value unless there's a natural order
- Up to about 12 bars; group the tail into "Other"
- No group under 10 people

1. **Insight headline**
   Names the two families and the benchmark they beat.
2. **Subtitle**
   Metric, unit, scope, period.
3. **Focus in blue**
   Only the bars the headline names; two at most.
4. **Echo in the label**
   Focus categories in bold navy.
5. **Values on the bars**
   So the value axis and gridlines go.
6. **Labelled benchmark**
   A thin dashed line, named where it sits.
7. **Zero baseline**
   A thin axis line; bars always start at zero.
8. **Grey context**
   Everything else, sorted by value.
9. **Footnote and source**
   Defines the metric; names system, date, population.



MAGNITUDE

Bar chart variants

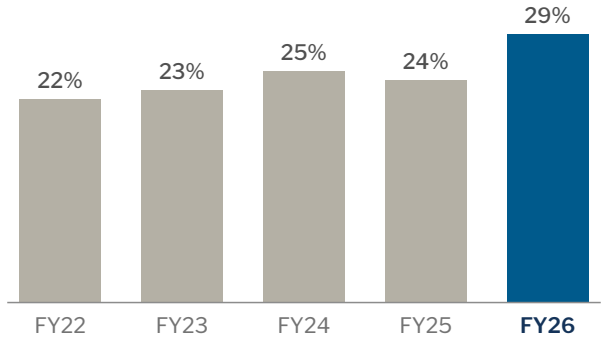
Same rules, different shapes. Pick the variant from the comparison you need to make.

When a bar is the wrong chart

- Values bunched far from zero (engagement 71–78): use a **dot plot**
- Largest more than 20× the smallest: use **proportional bubbles** or split the chart
- Parts of a whole: use a **100% stacked bar** or **waterfall**
- Many periods: use a **line**
- Spread across people: use a **distribution**

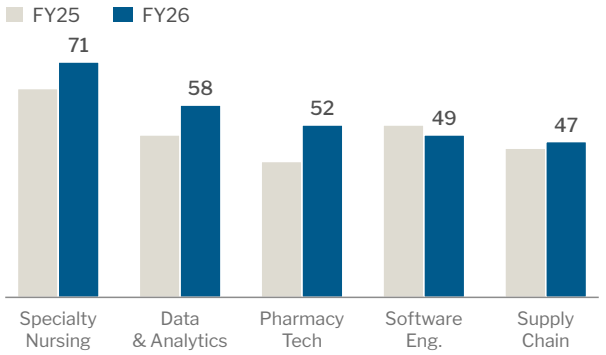
Column chart

A few periods in time order, or short category names. Here internal fill rose from 22% to 29%.



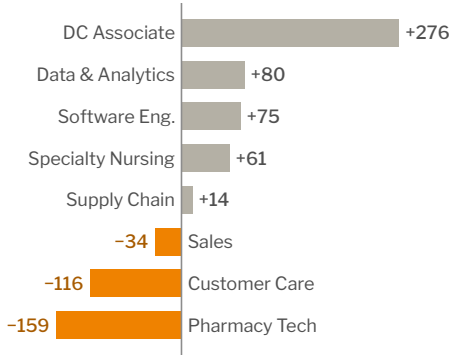
Grouped bars

Two or three series per category. Prior period in light grey, current in blue; key top-left.



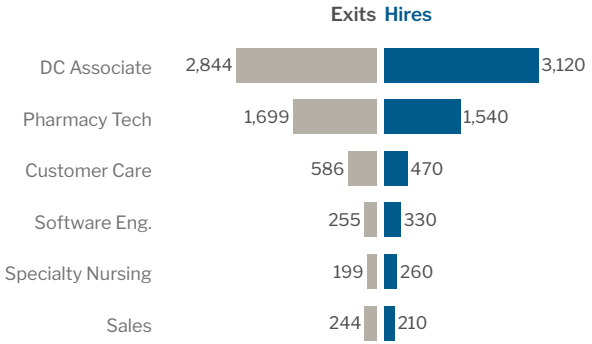
Diverging bars

Values on both sides of zero. Three job families shrank in FY26 despite hiring; negatives in orange.



Butterfly chart

Two measures back to back on the same items: FY26 exits and hires by job family.



MAGNITUDE

Bar chart best practice

Four rules that fix most bar charts.

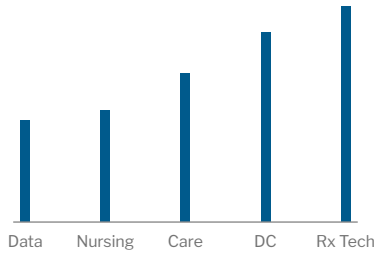
Size of bars. Thin bars pull the eye to the empty space between them. Make each bar about twice as wide as the gap.

Color of bars. A different color per bar distracts and implies categories that don't exist. One color; shade only the highlight.

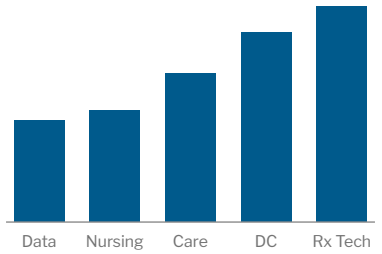
Direct labels. Gridlines and a value axis add noise. Label the bars and delete both.

Zero baseline. Bar length encodes quantity, so never truncate. If the data needs a narrow range, switch to a line or dot plot.

Size of bars

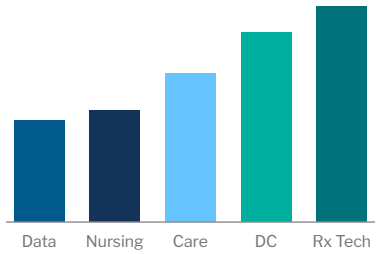


Try to avoid
Stick-thin bars; the gaps dominate.

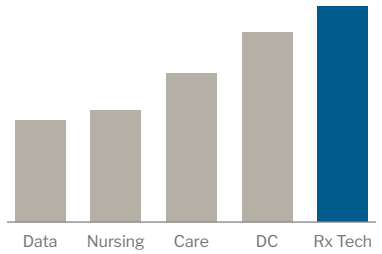


Recommended
Bar about twice the gap.

Color of bars

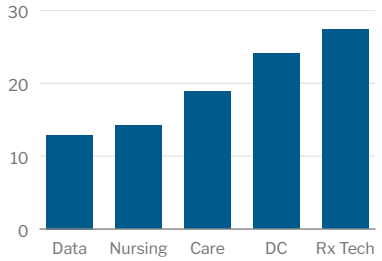


Try to avoid
A color per bar, encoding nothing.

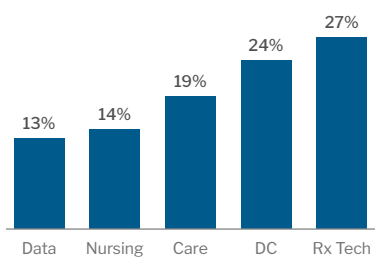


Recommended
Grey bars; blue for the one that matters.

Direct labels

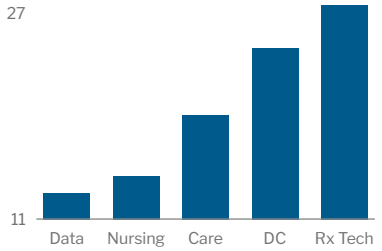


Try to avoid
Gridlines and an axis to read values from.

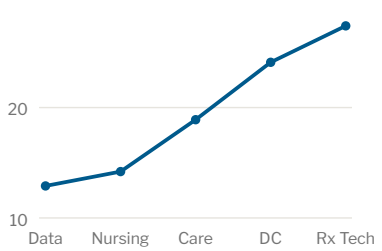


Recommended
Values on the bars; no axis, no grid.

Zero baseline



Try to avoid
A truncated axis exaggerates differences.



Recommended
A line or dot plot can use a narrow range.

Bar chart best practice, continued

Four more, from reviewing workforce decks.

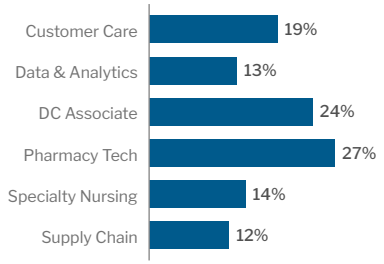
Sort with intent. Alphabetical order hides the ranking. Sort by value unless the categories have a natural order: tenure bands, job levels, survey scales.

Horizontal for long names. Rotated labels are slow to read. Job-family names need horizontal bars.

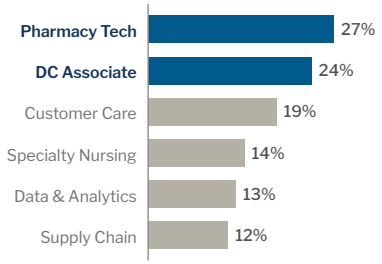
Label the benchmark in place. A reference line explained in a legend makes the reader search. Name it on the line.

Two highlights at most. Five colored bars in two colors is decoration, not emphasis.

Sort with intent

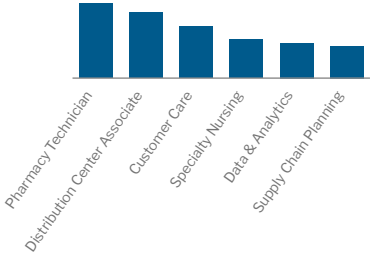


Try to avoid
Alphabetical: the ranking is hidden.

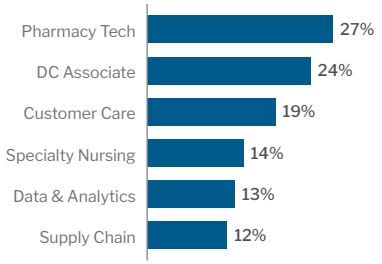


Recommended
By value, largest on top.

Horizontal for long names

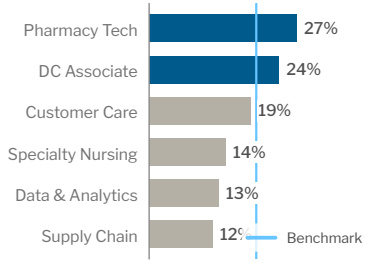


Try to avoid
Tilted names the reader has to decode.

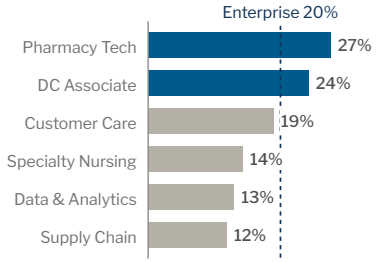


Recommended
Horizontal bars; names read naturally.

Label the benchmark

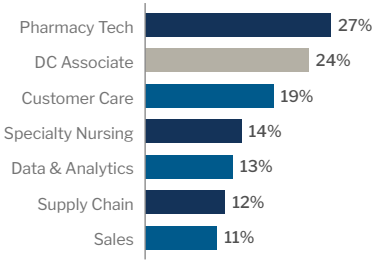


Try to avoid
A legend for one line, in a highlight color.

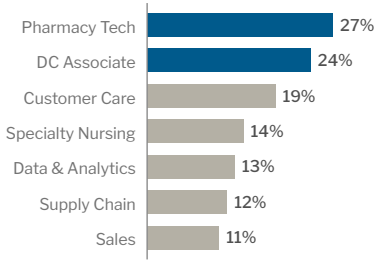


Recommended
Navy dashed line, named on the line.

Two highlights at most



Try to avoid
Everything highlighted means nothing is.



Recommended
The two the headline names.

Build a bar chart

The same chart in every tool we use. Each recipe starts from the starter kit, so the colors and fonts are already right.

Full, tested code for each platform is on the cookbook page: docs/cookbook/bar.md.

Before you send it

- [ ] Starts at zero; bar about 2× the gap
- [ ] Sorted by value or a natural order
- [ ] Values labelled; no value axis or gridlines
- [ ] At most two highlights, matching the headline
- [ ] Benchmark labelled in place
- [ ] No group under 10 people

PowerPoint and Excel	Start from the starter deck or chart workbook in `dist/office/`. Insert a clustered bar, sort the data ascending (so the largest is on top), set **Gap Width 50%**, fill all bars with Context grey and the focus bars with Focus blue. Add data labels at Outside End; delete the value axis, gridlines, legend and chart title. The slide title is the headline.
Power BI	Import `mckesson-dataviz-theme.json` and sort the bars by the measure. Drive the highlight from a measure, not by clicking bars: a `Bar color` measure returns the focus blue when attrition is above 20% and context grey otherwise, applied through **Colors › fx › Field value**. Turn on data labels; turn off the X axis and gridlines. See docs/powerbi-setup.md.
Python	matplotlib with `mckviz`; Plotly uses the same helpers (`mv.pl_highlight`).

```
bars = ax.barh(df.job_family, df.rate, height=0.67,
               color=mv.highlight(df.job_family, focus=focus))
mv.label_bars(ax, bars, fmt="{:.1f}%")
mv.clean_axes(ax, keep=("left",), grid=None); ax.set_xticks([])
mv.titles(fig, headline, subtitle); mv.source(fig, source)
mv.lint(fig)   # checks colors, zero baseline, labels vs. axis
```

HTML	Copy `templates/html/chart-card.html`: headline, subtitle and source are HTML, colors come from CSS variables such as `var(--mck-chart-focus)`, and the chart is drawn after the font loads.
R	`source("dist/r/theme_mck.R")`, then `geom_col(aes(fill = job_family), width = 0.67)` + `scale_fill_manual(values = mck_highlight(levels(df$job_family), focus = focus), guide = "none")` + `theme_mck()`.
Coding agents	"Make a McKesson bar chart of voluntary attrition by job family, FY26, highlighting the families above the enterprise rate." The `mck-chart` skill runs the workflow and returns the chart, a chart brief and alt text.

PART-TO-WHOLE

Stacked bar chart

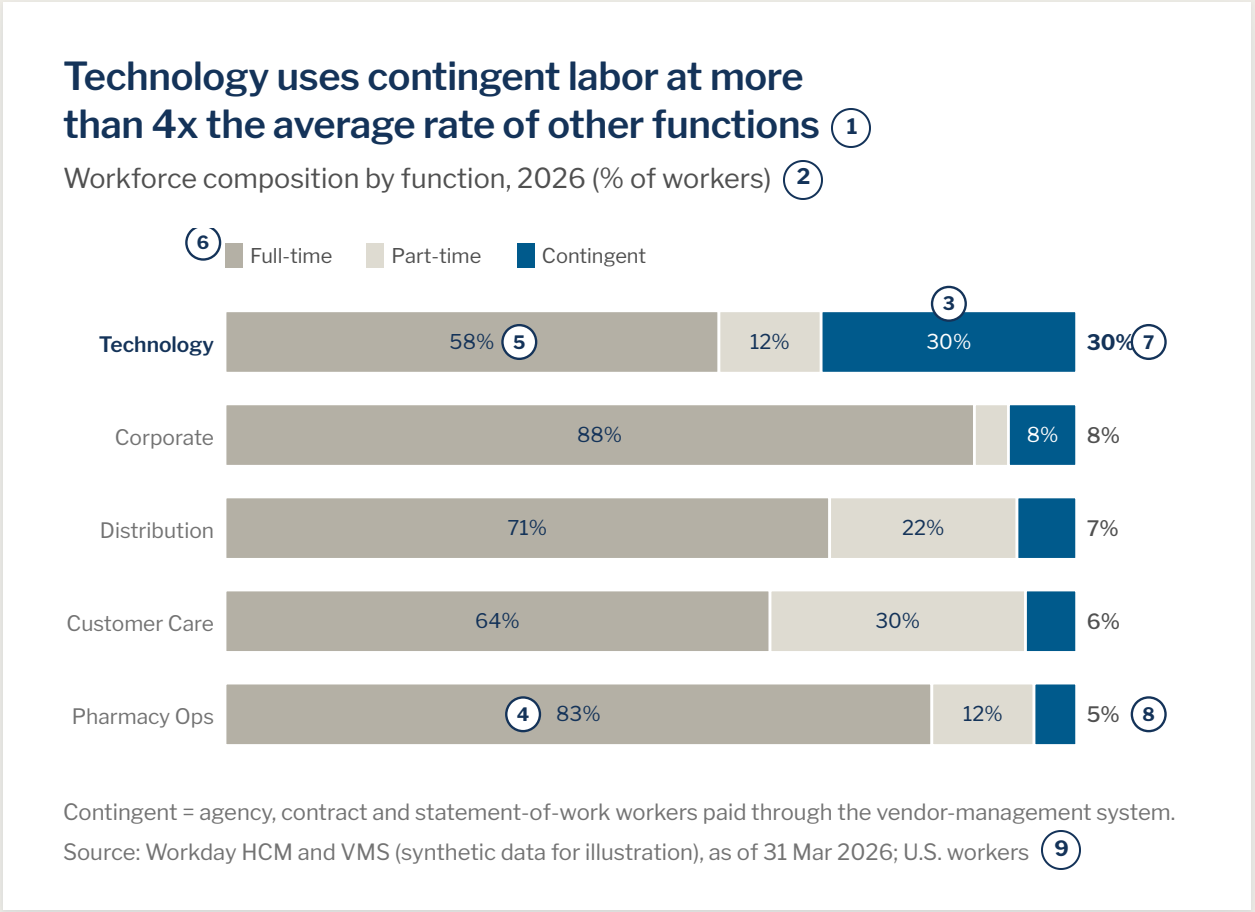
Part-to-whole charts show how a whole breaks into its parts, and how each part relates to the total.

A 100% stacked bar compares the mix of several wholes at once. Color only the segment the headline is about, and put it on the baseline or at the end of each bar.

McKesson spec

- Five segments at most; group the rest as "Other"
- White separators between segments
- Labels inside segments that fit; key top-left in stack order
- Sort the bars by the segment that matters

1. **Insight headline**
   Names the part and how far it stands out.
2. **Subtitle**
   Metric, unit, scope, period.
3. **The part that matters**
   One segment in blue, at the end of every bar.
4. **Grey context**
   Two greys for the other parts.
5. **Labels in the segments**
   Where they fit; navy on grey, white on blue.
6. **Key in stack order**
   Top-left, left to right as the segments run.
7. **Echo the value**
   The story segment's value at the bar end, bold for the focus bar.
8. **Sorted by the story**
   Bars ordered by the blue segment, not alphabetically.
9. **Footnote and source**
   Define the parts.



PART-TO-WHOLE

Stacked bar variants

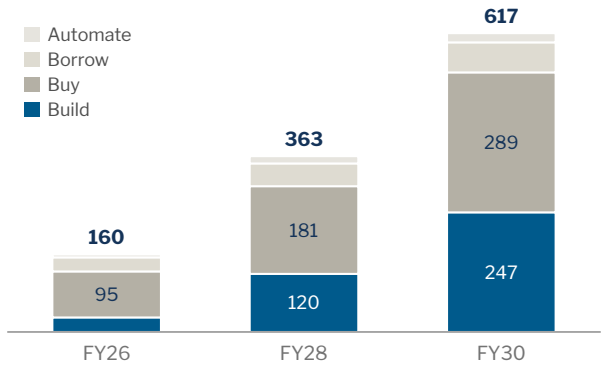
Same rules: one part in color, the rest grey, labels in place.

When a stack is the wrong chart

- One whole with up to five parts: a **donut** or a single bar
- How a total changes step by step: a **waterfall**
- Many parts or a hierarchy: a **treemap**
- Comparing the parts themselves across groups: **grouped bars** or **small multiples**; only the baseline segment of a stack compares cleanly

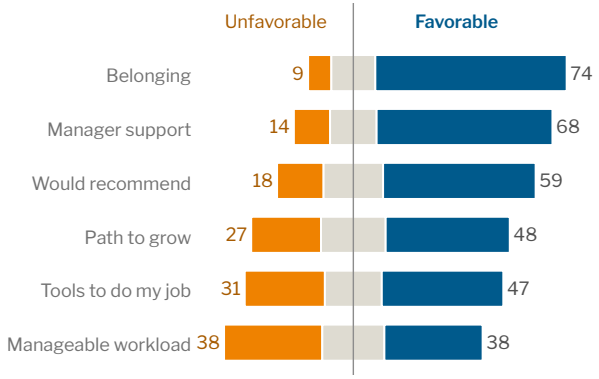
Stacked columns with totals

When the total matters too. The FY30 Data & AI gap of 617 FTE: reskilling (on the baseline) closes 40%.



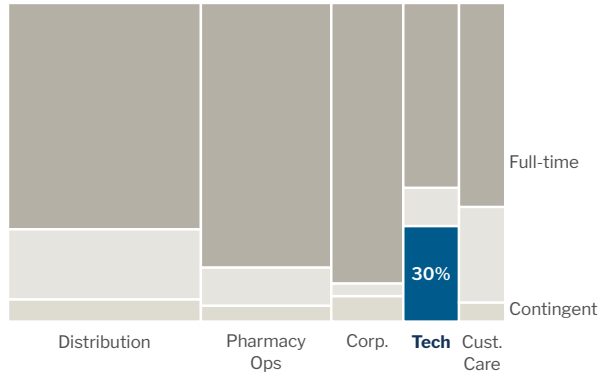
Diverging stacked bars

Survey scales. Unfavorable in orange left of zero (it means worse), favorable in blue, neutral split across the middle.



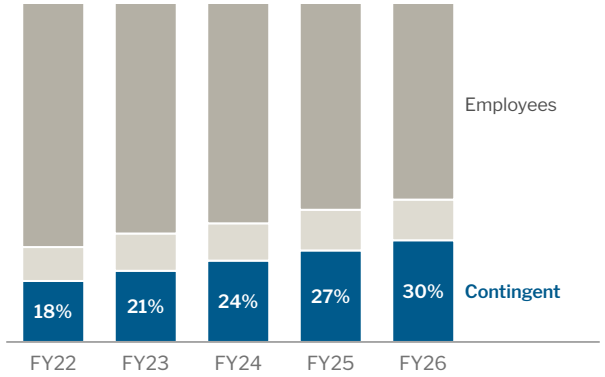
Marimekko

Width and height both carry data: column width is each function's workforce, height its mix. Powerful, but it needs explaining.



Mix over time

A few periods as 100% columns, the tracked segment on the baseline so its change reads directly.



PART-TO-WHOLE

Stacked bar best practice

Stacks are easy to make and easy to misread. Four rules keep them honest.

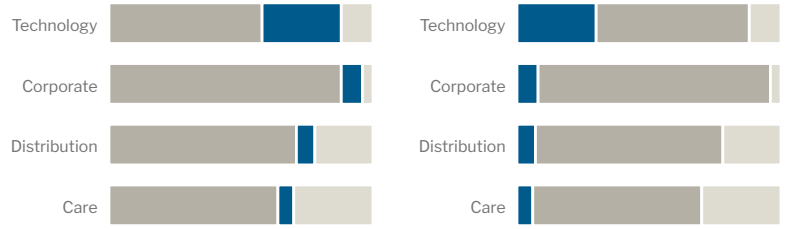
The part that matters on the baseline. Only segments that share a starting line can be compared by eye.

Five segments at most. More slices turn a bar into a barcode. Keep the parts the story needs and group the rest.

Label in place; key in stack order. If a key is needed, put it top-left, in the same order as the segments.

One order in every bar. Never re-sort segments within each bar.

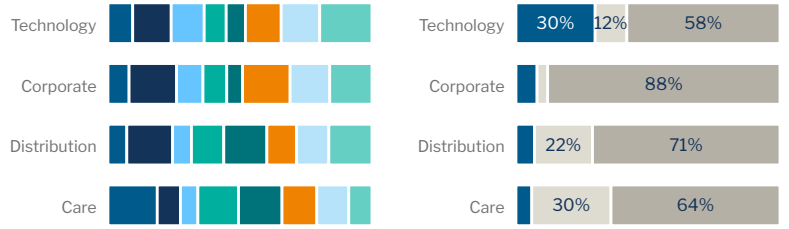
The part that matters on the baseline



Try to avoid
A floating segment can't be compared across bars.

Recommended
On the baseline, every bar starts level.

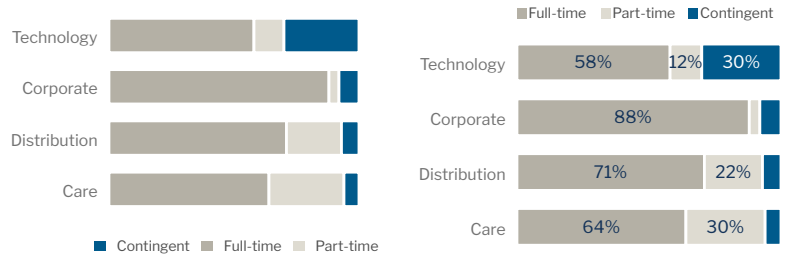
Five segments at most



Try to avoid
Eight colored slices nobody can follow.

Recommended
The part in blue, the rest in grey.

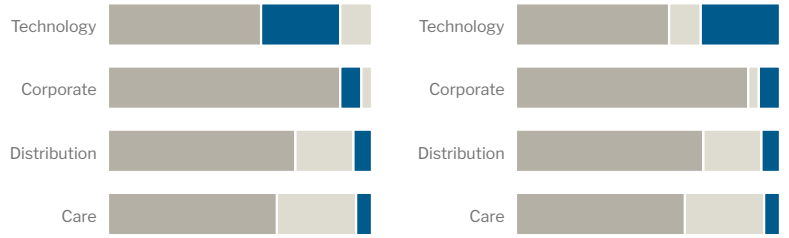
Label in place, key in order



Try to avoid
A bottom legend in a different order.

Recommended
Values inside, key top-left in stack order.

One order in every bar



Try to avoid
Segments re-sorted inside each bar.

Recommended
The same order everywhere.

PART-TO-WHOLE

Waterfall chart

A waterfall shows how a starting total becomes an ending total, one driver at a time.

It is a stacked column chart with an invisible spacer series that floats each driver at the running total. Use it for labor cost bridges, headcount bridges and budget-to-actual walks.

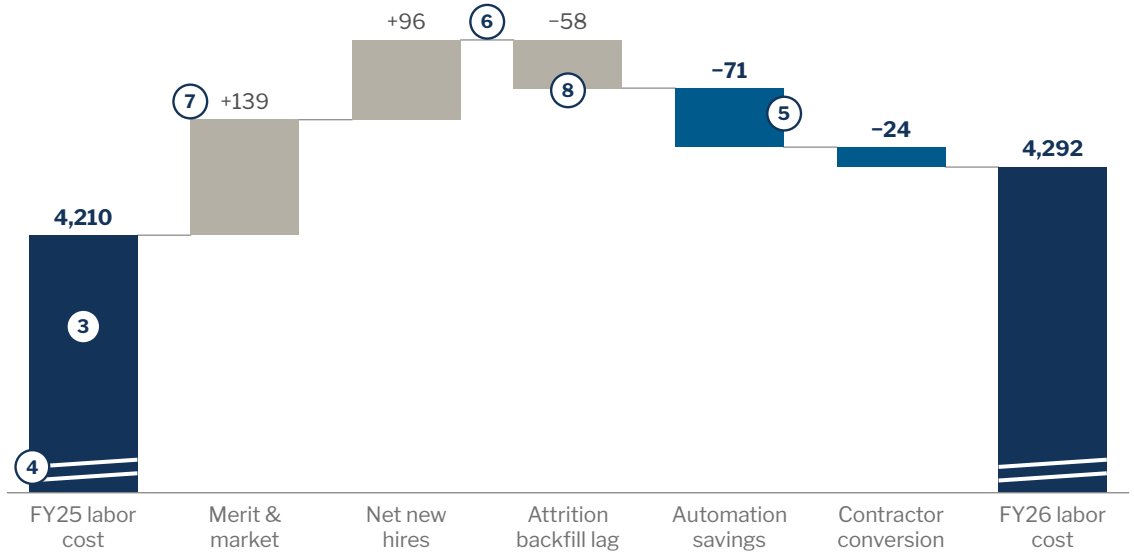
McKesson spec

- Totals in navy; drivers grey; the drivers the headline names in blue
- Thin light connectors between bars
- Signed labels on drivers (true minus, explicit +); bold totals
- If the axis starts above zero, mark the break and say so

1. **Insight headline**
   Sizes what the named drivers did to the total.
2. **Subtitle**
   From and to, with the unit.
3. **Totals in navy**
   Start, end and any subtotal.
4. **Marked break**
   The axis starts at $3,900M; white slashes say so.
5. **The drivers that matter**
   In blue; label bold.
6. **Thin connectors**
   Light grey, so the bars stay dominant.
7. **Signed labels**
   +139, −58: direction without color.
8. **Grey context**
   Every other driver, in logical order.
9. **Footnote**
   States the axis break.

Automation and contractor conversion offset ~40% of FY26 labor cost growth (1)

Labor cost bridge, FY25 to FY26 ($M) (2)



Value axis starts at $3,900M (break marks on the totals) so the drivers are readable.

Source: Finance labor cost model (synthetic data for illustration), FY26 (9)

PART-TO-WHOLE

Waterfall variants

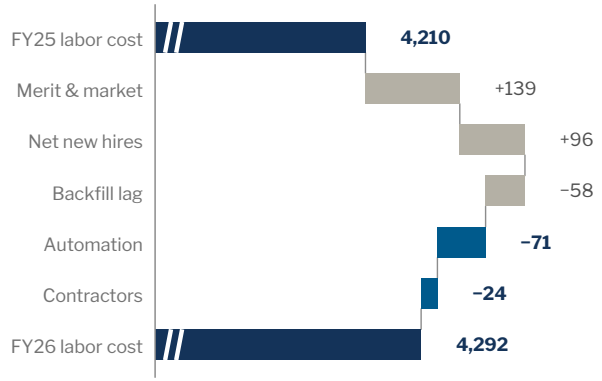
The same build, turned or recolored for the job.

Color by direction only when direction is the point. In a headcount bridge, additions in blue and losses in orange read instantly, because orange means "worse". In a cost bridge the default stays grey with a blue highlight.

Excel's built-in Waterfall chart is vertical only. For horizontal waterfalls or per-bar colors, build a stacked bar with a transparent spacer series (see the cookbook).

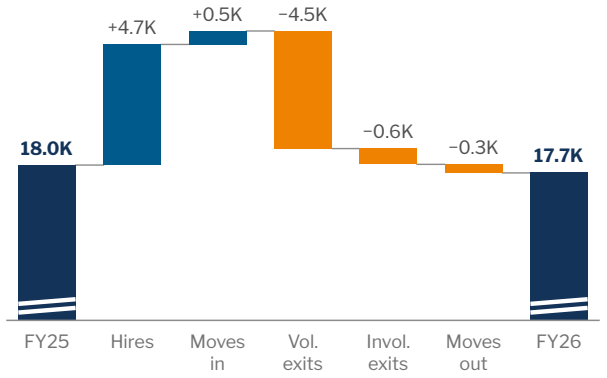
Horizontal waterfall

Long driver names, narrow layouts and phones.



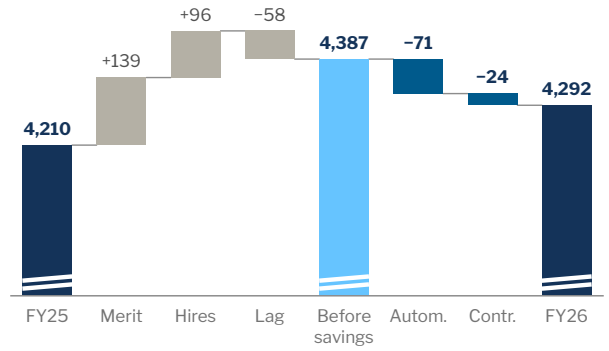
Colored by direction

Frontline headcount bridge: voluntary exits equal 97% of external hires.



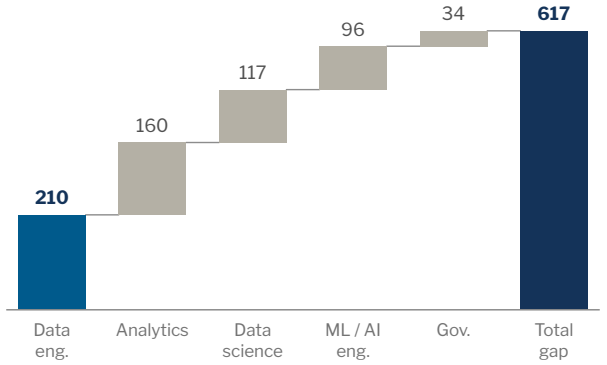
With a subtotal

A light-blue subtotal separates cost growth from the savings that offset it.



Distribution waterfall

Contributions to one total instead of steps in time: data engineering is 34% of the FY30 gap.



PART-TO-WHOLE

Waterfall best practice

The bars carry the story; everything else stays quiet.

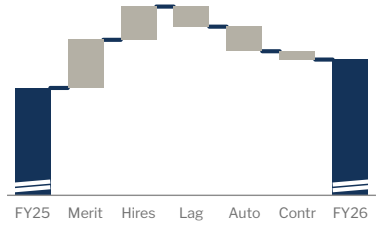
Thin connectors. Heavy dark connectors compete with the bars. Use hairlines in light grey.

No red and green. They fail for color-blind readers and aren't McKesson colors. Grey drivers with a blue highlight, or blue and orange when direction is the story.

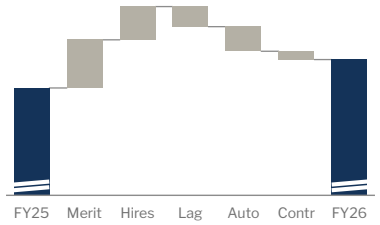
Mark a broken axis. Totals often dwarf the drivers. Starting the axis above zero is allowed here, and only here, if the break is drawn and footnoted.

Group the small steps. Keep the drivers the story needs; fold the rest together.

Connector lines

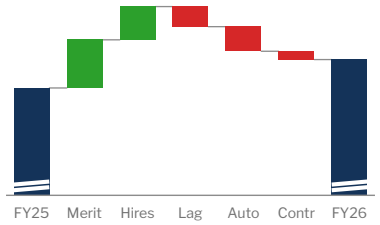


Try to avoid
Heavy, dark connectors.

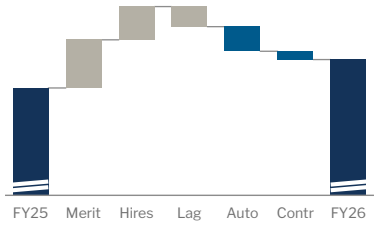


Recommended
Thin, light grey.

Coloring the bars

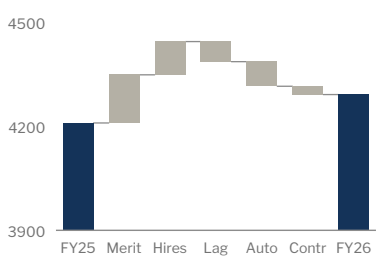


Try to avoid
Green up, red down.

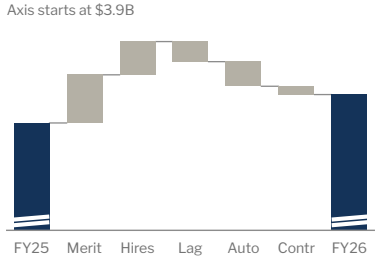


Recommended
Grey drivers; the named ones blue.

A broken axis

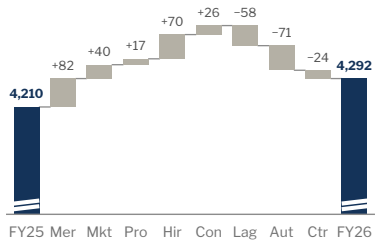


Try to avoid
Cut totals with no warning.

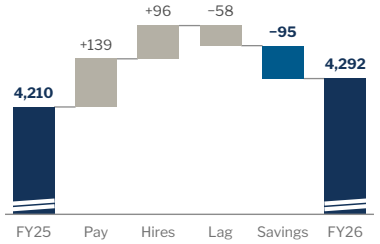


Recommended
Break marks and a note.

Group the small steps



Try to avoid
Ten narrow drivers to decode.



Recommended
Five steps, savings combined.

PART-TO-WHOLE

Treemap

Nested rectangles whose area is proportional to value. A treemap shows many parts, or a hierarchy, in one compact block.

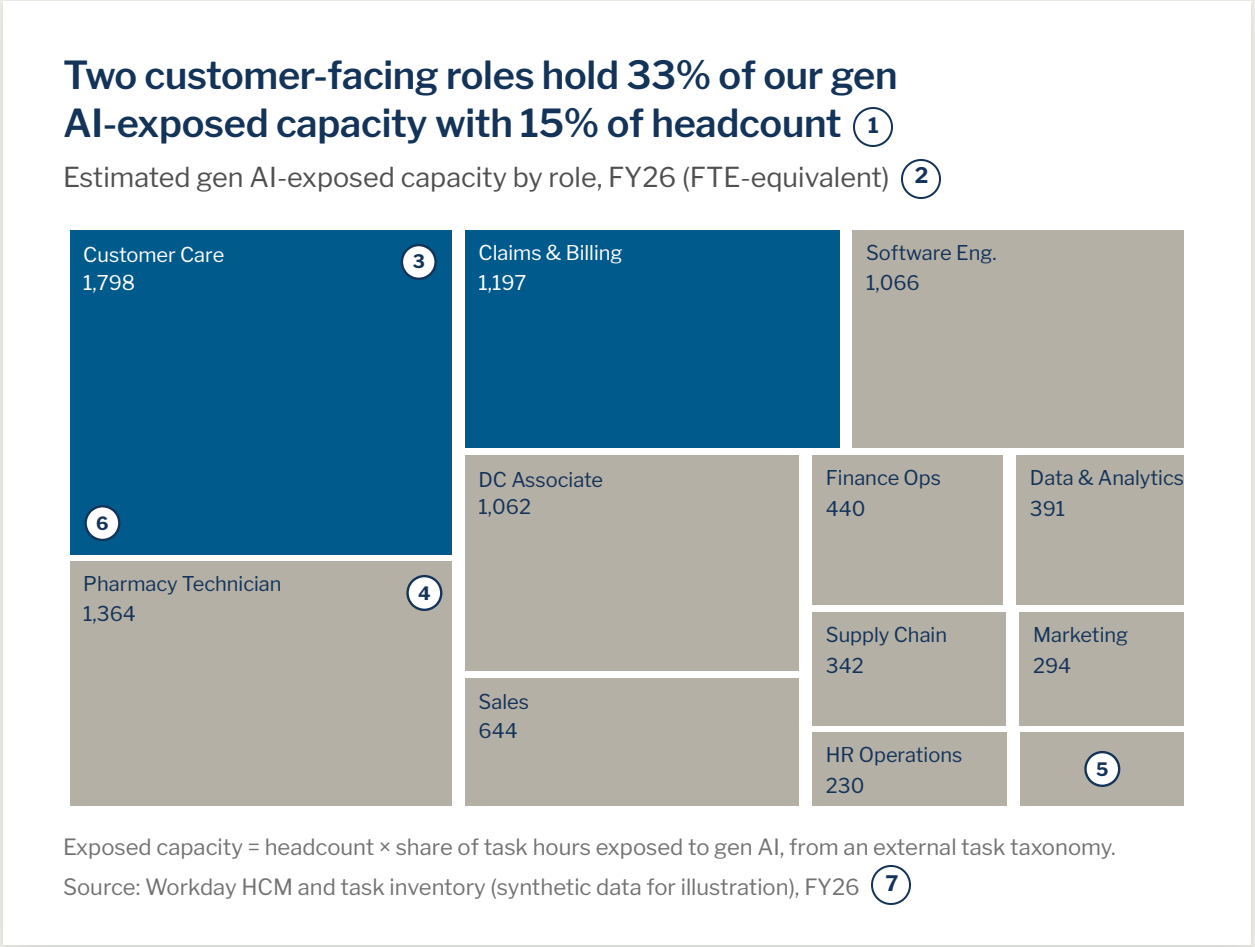
Use it when there are too many parts for a donut or a stack (ten to thirty roles, sites or skills) and the point is which few dominate.

Don't use it when readers need to compare similar sizes precisely, or there are only a handful of parts: a sorted bar is clearer.

McKesson spec

- Squarified layout, largest top-left
- Thin, even white gutters
- Grey tiles; the ones the headline names in blue
- Name and value inside each tile that fits

1. **Insight headline**
   Share of the whole held by the named few.
2. **Subtitle**
   What area measures, with the unit.
3. **Focus in blue**
   The two roles the headline names.
4. **Grey context**
   Every other role, the same grey.
5. **Smallest last**
   Bottom-right. Tiles too small for a label get none; list them in a footnote if they matter.
6. **Largest first**
   Top-left, flowing to the smallest; labels top-left inside each tile.
7. **Footnote and source**
   Define the measure.



PART-TO-WHOLE

Treemap best practice

A treemap works when it reads as one shape that flows from large to small.

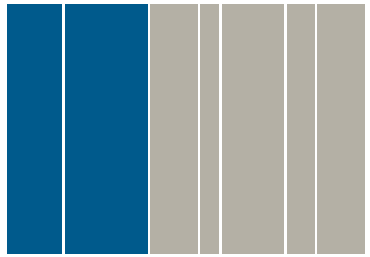
Largest to smallest. Don't arrange tiles alphabetically or by an org grouping that breaks the flow.

Labels inside. No label bands or boxes over the tiles; they distort the areas. If a tile can't hold its label, leave it unlabelled or label it outside.

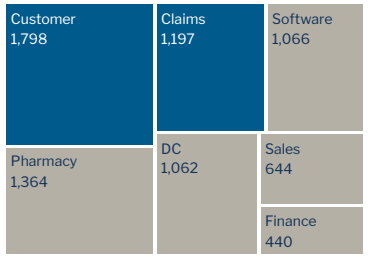
Even padding. Uneven gaps make similar tiles look different.

Few parts? Use a bar. Areas are harder to compare than lengths.

Arrange by size

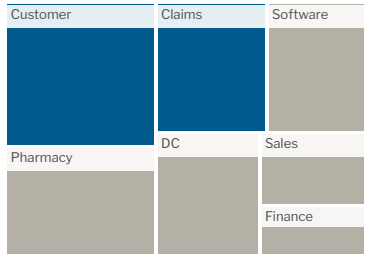


Try to avoid
Alphabetical strips.

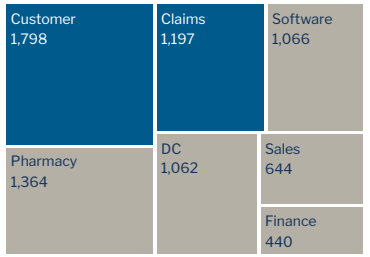


Recommended
Largest top-left.

Labelling



Try to avoid
Label bands over the tiles.



Recommended
Inside, top-left.

Padding

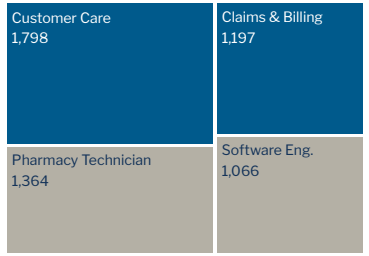


Try to avoid
Uneven gaps.

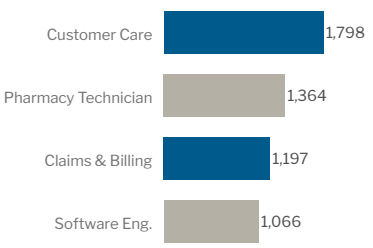


Recommended
Thin, even gutters.

Few parts



Try to avoid
Four similar tiles.



Recommended
A sorted bar ranks them.

PART-TO-WHOLE

Donut and pie

A donut shows one whole split into a few parts. Prefer it to a pie: readers follow the arc length around the ring, and the center holds the total.

Use it for one whole and at most five parts. To compare several wholes, use 100% stacked bars; for more parts, a bar or treemap.

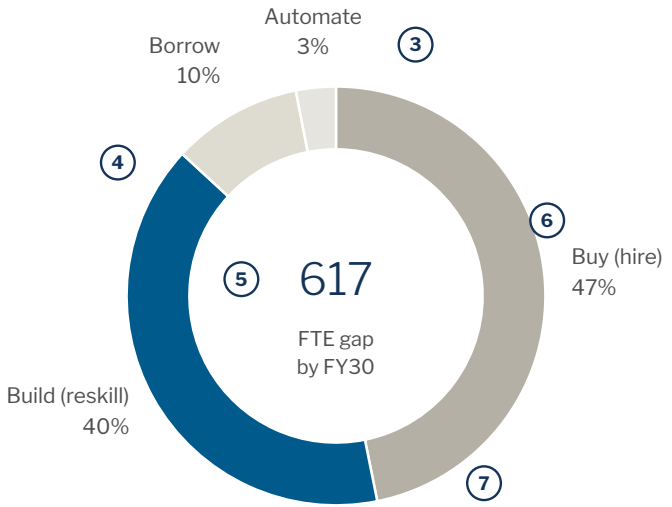
McKesson spec

- Five segments at most; "Other" last
- Largest first, clockwise from 12 o'clock
- Hole about 60% of the diameter; white separators
- Labels and values outside; no legend

1. **Insight headline**
   The share, and what it means for the plan.
2. **Subtitle**
   The whole, the unit, the year.
3. **Start at 12**
   Largest first, then clockwise.
4. **The part that matters**
   In blue; the rest grey.
5. **Total in the center**
   Light weight, large.
6. **Labels outside**
   Name and share, beside each segment.
7. **White separators**
   Thin gaps keep segments distinct.
8. **Footnote and source**
   Define the parts.

Reskilling can close 40% of the FY30 Data & AI gap; hiring still has to cover nearly half (1)

Planned sources to close the Data & AI capability gap, FY30 (% of 617 FTE) (2)



Base scenario. Build = reskilling current employees; buy = external hires; borrow = contractors and partners; automate = roles removed by automation.

Source: Strategic workforce planning model (synthetic data for illustration), Mar 2026 (8)

PART-TO-WHOLE

Donut best practice

People judge length faster than angle or area, which is why the donut beats the pie.

Donut over pie. The ring turns angles into arc lengths, and the hole holds the total.

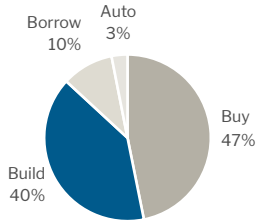
Labels outside. Squeezed labels on the ring are hard to read; put names and values outside.

Few segments. Group the smallest into "Other".

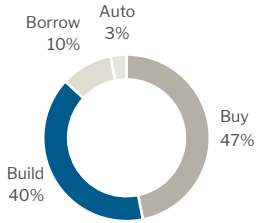
Largest first, clockwise from 12. Don't mix small and large at random.

Progress rings and dials for a single percentage are in Single number.

Donut over pie

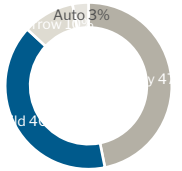


Try to avoid
Angles and areas.

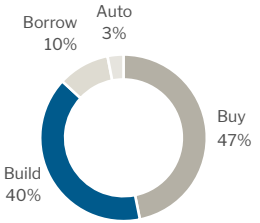


Recommended
Arc lengths on a ring.

Labelling



Try to avoid
Labels squeezed onto the ring.

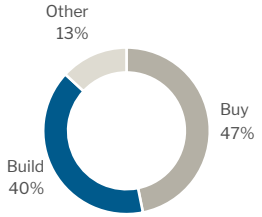


Recommended
Outside, with the share.

Many small segments

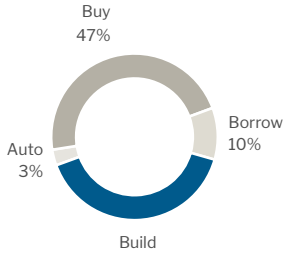


Try to avoid
Nine slices, nine colors.

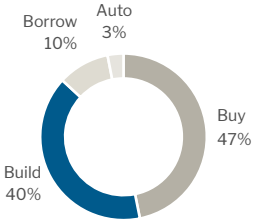


Recommended
The story, plus "Other".

Order of segments



Try to avoid
No order, odd start.



Recommended
Largest first from 12.

PART-TO-WHOLE

Build part-to-whole charts

Every tool can make these natively. The settings below are the ones that matter; tested code and click-by-click steps are on the cookbook pages.

stacked-bar.md · waterfall.md · treemap.md · donut-pie.md

Before you send it

- [ ] Parts sum to the whole (100%, or the end total)
- [ ] Five parts at most; "Other" last
- [ ] One part in color, matching the headline
- [ ] Waterfall axis break marked and footnoted
- [ ] No group under 10 people, in any part

PowerPoint and Excel	**Stacked:** Stacked or 100% Stacked Bar, Gap Width 50%, Overlap 100%, white 1.5 pt borders; move the story series first or last in Select Data. **Waterfall:** Insert › Waterfall, then right-click each total › *Set as Total*; drivers grey, totals navy. **Treemap:** Insert › Hierarchy › Treemap; labels *Overlapping*, not Banner. **Donut:** Doughnut, angle of first slice 0°, hole 62%, data sorted descending.
Power BI	Native 100% stacked bar, Waterfall, Treemap and Donut visuals all take the theme. Set the story color per series or with a field-value color measure; legend top-left or off; data labels on. The native waterfall can't color single bars or break the axis: use a stacked column with a transparent spacer measure when you need that.
Python	`ax.barh(..., left=running_total, edgecolor="white")` for stacks; the waterfall, squarified treemap and donut recipes in the cookbook use only matplotlib and `mckviz`. Plotly has native `go.Waterfall`, `go.Treemap` and `go.Pie(hole=0.6)`.
HTML	Plotly with `MCK_PLOTLY_TEMPLATE`: `barmode: "stack"`, `waterfall` and `treemap` traces. Read colors from `var(--mck-chart-focus)` and `var(--mck-chart-context)`.
R	`geom_col(position = "fill")` for 100% stacks; `treemapify::geom_treemap()` for treemaps; waterfalls with `geom_rect()`. Colors from `mck_highlight()`.

```
colors = mv.highlight(parts, focus="Contingent")          # one part blue, rest grey
mv.lint(fig)    # flags > 6 colors, legends, and bars that don't start at zero
```

CHANGE OVER TIME

Line chart

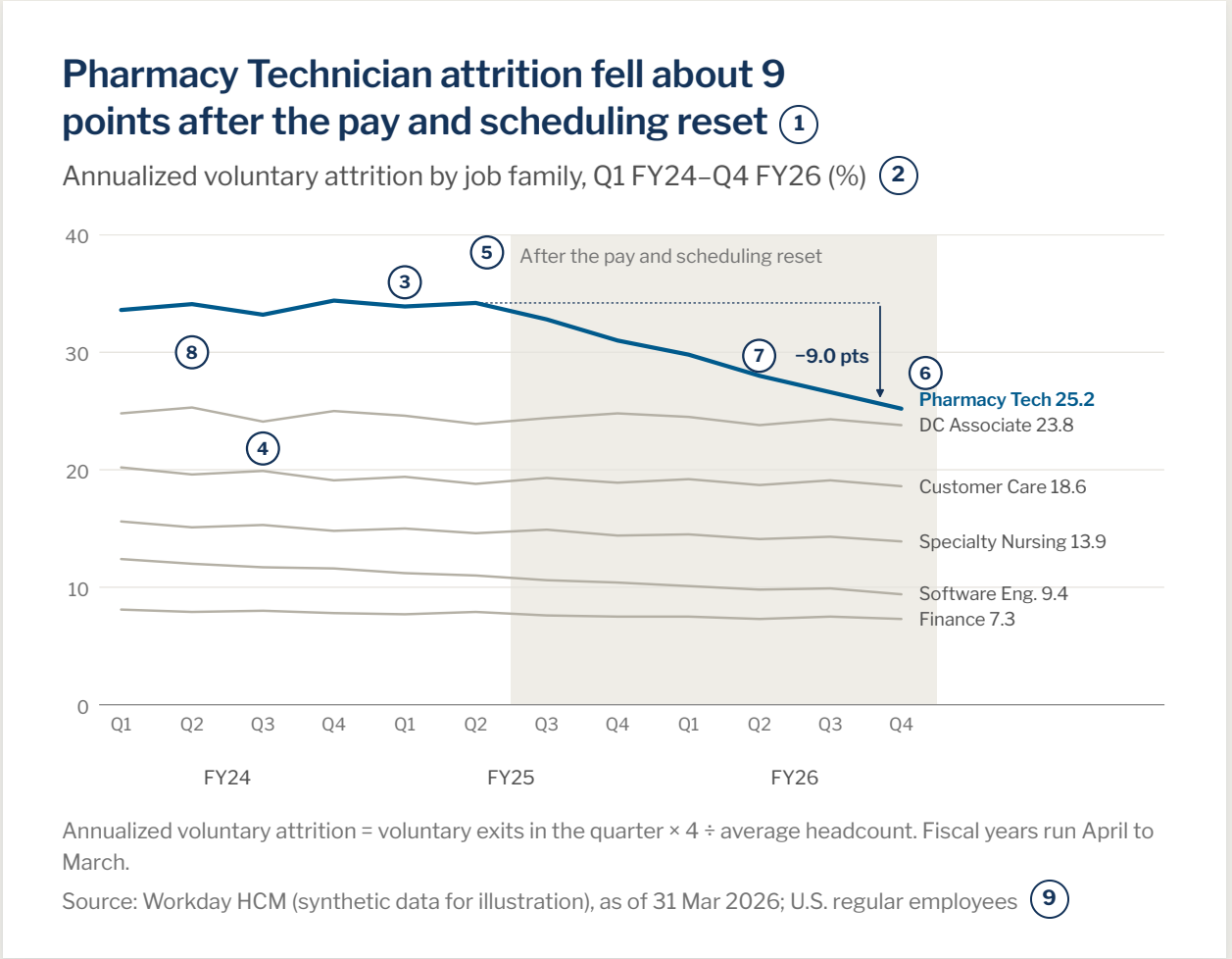
Change-over-time charts show how something moves, and when it changed, at any time scale from weeks to decades.

A line connects values at equal time intervals, so the eye reads the trend rather than the individual amounts. Use it for monthly or quarterly attrition, hiring velocity, or a forecast.

McKesson spec

- The series the headline names in blue; others grey
- Data lines about twice the gridline weight
- Names at the line ends, not in a legend
- Shade the period that explains the change

1. **Insight headline**
   What changed, by how much, and why.
2. **Subtitle**
   Metric, unit, scope and period.
3. **Focus line**
   Blue and heavier than the rest.
4. **Context lines**
   Grey and thinner: there to compare against.
5. **The period that matters**
   Lightly shaded and named.
6. **Direct labels**
   Names and latest values at the line ends, nudged apart.
7. **The change, sized**
   A short callout carries the headline number.
8. **Light scaffolding**
   A few thin gridlines; the unit lives in the subtitle.
9. **Footnote and source**
   Define the metric.



CHANGE OVER TIME

Line chart variants

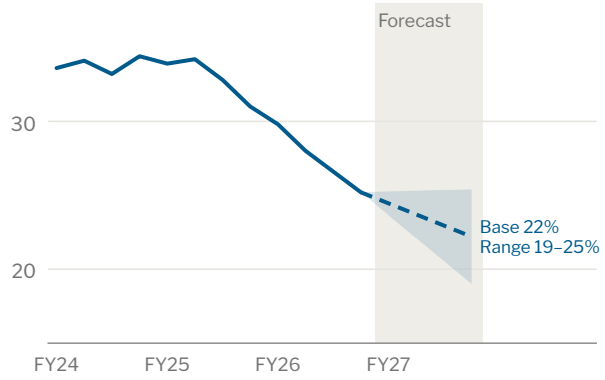
The same rules hold: one line in blue, names at the ends, light scaffolding.

When a line is the wrong chart

- Only two to five periods, and the amounts matter: **columns**
- How a total splits over time: **stacked area**
- Discrete events of different sizes: a **bubble timeline**
- Many series that tangle: **small multiples**
- Two measures in different units: two aligned charts, never a **dual axis**

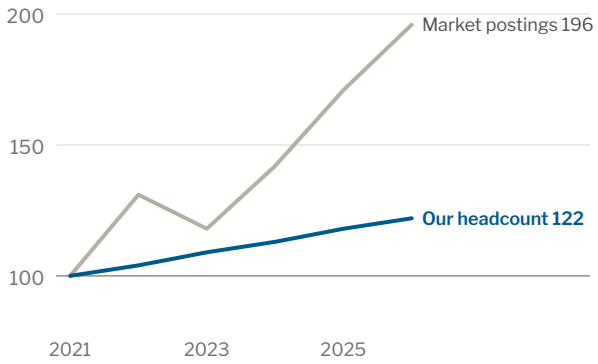
Forecast with a range

Dashed beyond the last actual, inside a band that widens with time. Label it as an estimate.



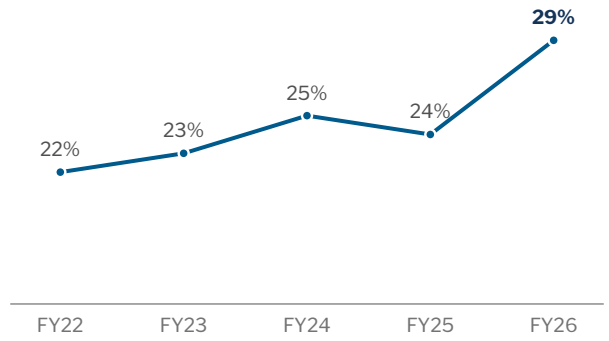
Indexed to 100

Compares growth when the units or sizes differ: market Data & AI postings reached 196, our headcount 122.



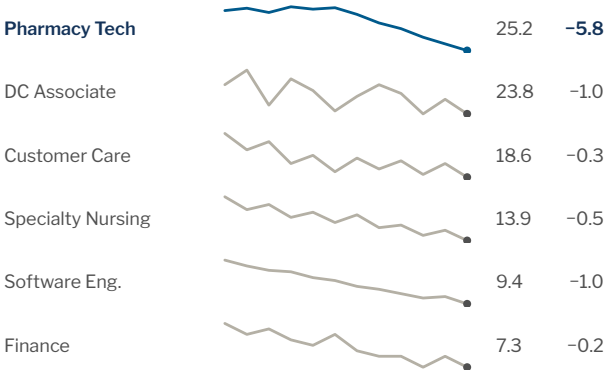
Few points, every value labelled

With five annual values, label each one and drop the axis.



Sparklines

One small line per row in a table or dashboard: shape, latest value, change. Worse in orange-brown.



CHANGE OVER TIME

Line chart best practice

Four rules that make trends readable.

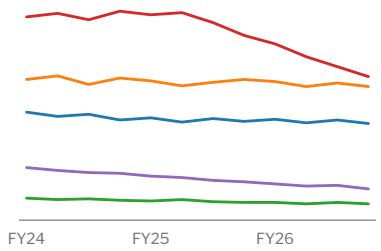
Highlight one line. A color per series doesn't show the trend; it makes the reader decode. Grey the background, color the message.

Lines for trends. Stacked columns hide how each part changes. When the trend is the point, draw lines.

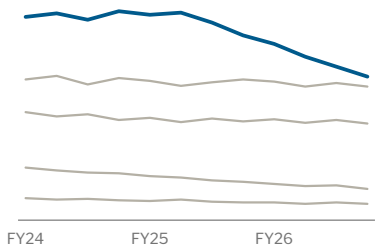
No fill without a zero. Shading under a line turns it into an area chart, which implies a zero baseline.

Names at the line ends. A legend above the chart breaks the reading order.

Highlights

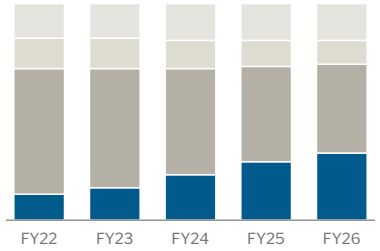


Try to avoid
Every line its own color.

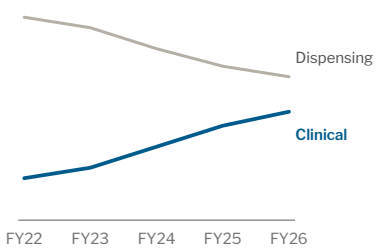


Recommended
Grey context, one blue line.

Showing trends

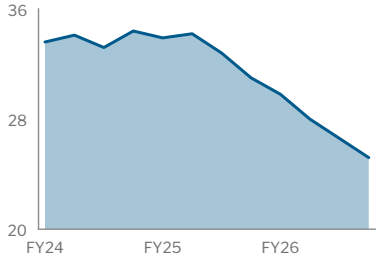


Try to avoid
Stacks hide each part's change.

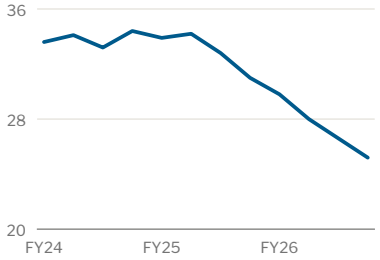


Recommended
Lines show the crossover.

Shading lines

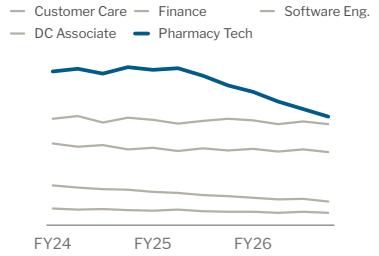


Try to avoid
A filled area on a 20–36 axis.

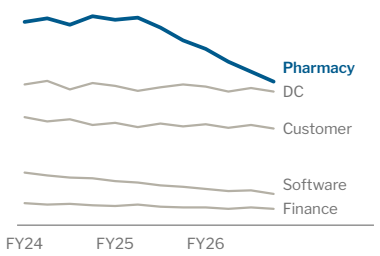


Recommended
A plain line.

Labelling lines



Try to avoid
A legend to decode.



Recommended
Names where the lines end.

CHANGE OVER TIME

Line chart best practice, continued

Four more, from HR and workforce data in particular.

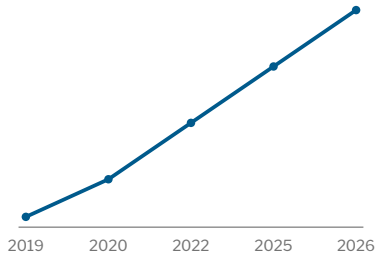
Honest time spacing. Missing years still take up space. Use a real date axis, not evenly spaced labels.

Show gaps as gaps. A system change or survey break is not a straight line. Leave the gap and say why.

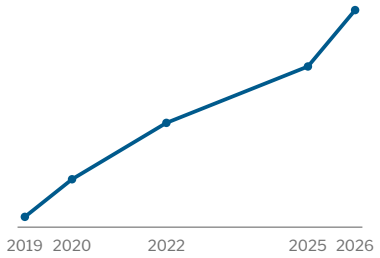
Weight with intent. Data lines about twice the gridlines; gridlines light, or gone.

Too many lines? Split them. More than four or five crossing lines become spaghetti. Use small multiples.

Even time intervals

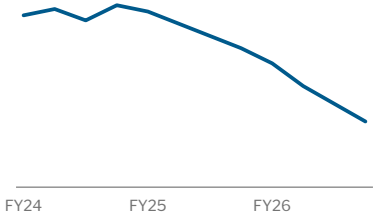


Try to avoid
Missing years squeezed out.

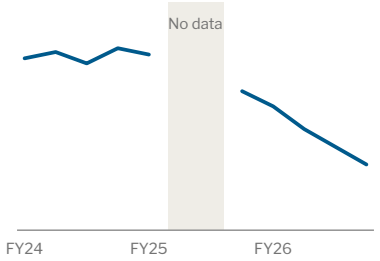


Recommended
True spacing; the slope is honest.

Missing data

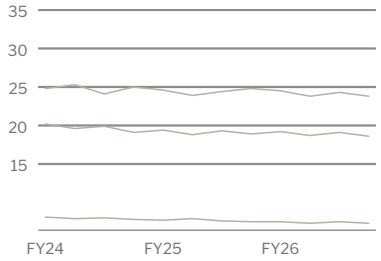


Try to avoid
Drawing across the gap.

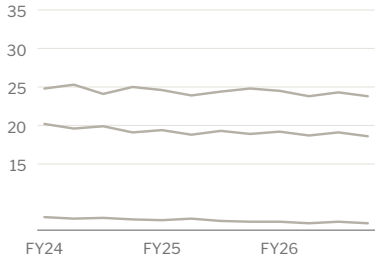


Recommended
A visible, labelled gap.

Line weight

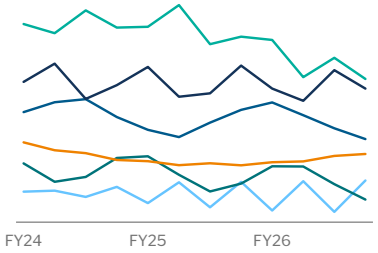


Try to avoid
Gridlines outweigh the data.

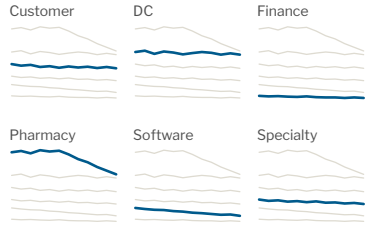


Recommended
Data lines twice the grid.

Too many lines



Try to avoid
Six crossing colors.



Recommended
Small multiples.

CHANGE OVER TIME

Stacked area chart

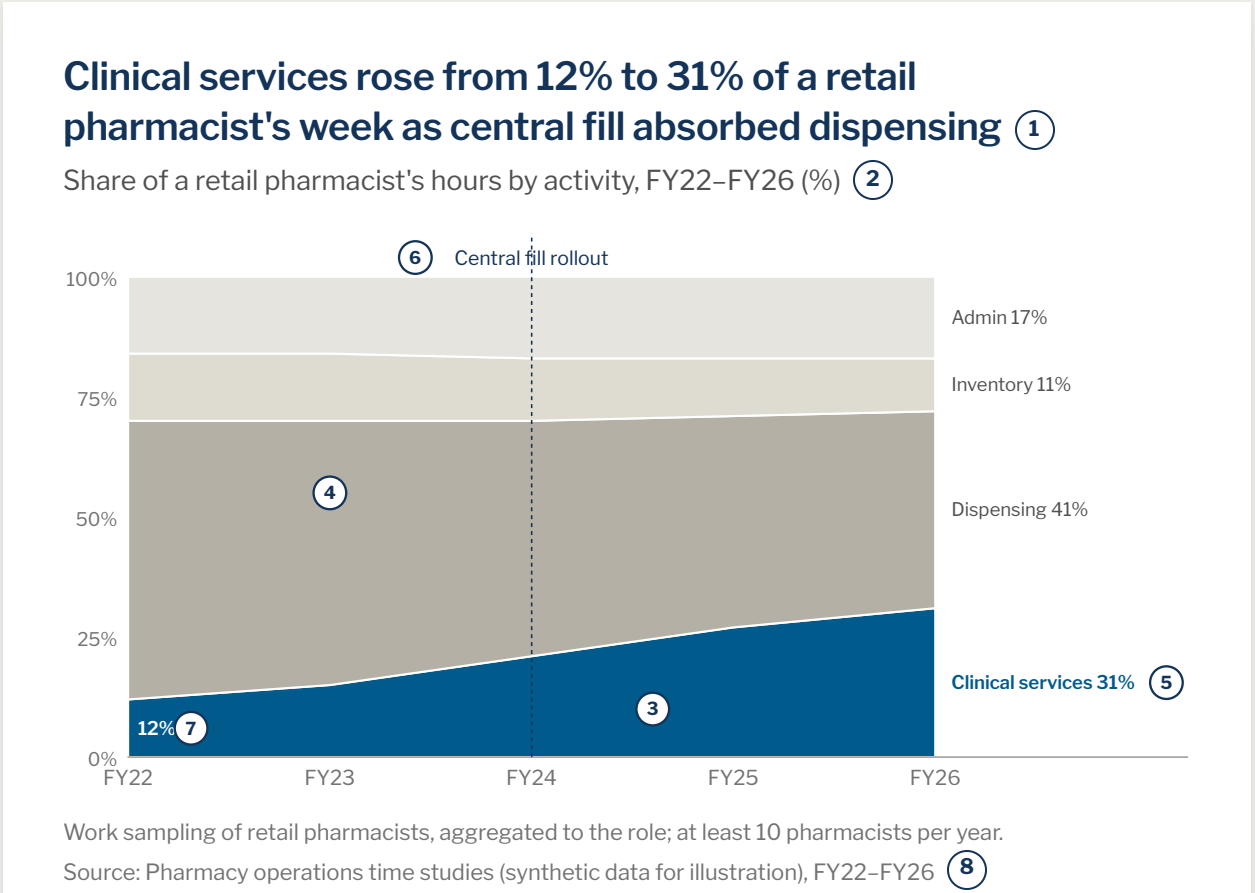
Layers stacked on each other show how a total splits over time, and how each part's share shifts.

Use it for a few parts that change gradually: how a role's hours divide between activities, or how the workforce mix moves. The layer on the baseline is the only one that reads precisely, so put the story there.

McKesson spec

- Four or five layers at most
- The story layer on the baseline, in blue; the rest ordered greys
- Labels inside or at the right edge, not in a legend
- Mark the event that explains the shift

1. **Insight headline**
   Start, end and the reason.
2. **Subtitle**
   Whose time, what unit, which years.
3. **Story on the baseline**
   Blue, so its growth reads directly.
4. **Grey context**
   Ordered from dark to light.
5. **Labels at the end**
   Name and latest share beside each layer.
6. **The event, marked**
   A thin dashed line and a short name.
7. **Where it started**
   The first value, so the change is explicit.
8. **Footnote and source**
   How time was measured.



CHANGE OVER TIME

Stacked area best practice

Areas are heavy marks. Keep them few, ordered and calm.

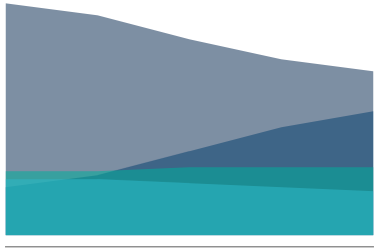
Stack, don't overlap. Areas drawn over each other hide what's behind. Stack them, or use lines.

Order the colors. Random colors make neighbors clash. Story layer blue on the baseline; the rest in ordered greys.

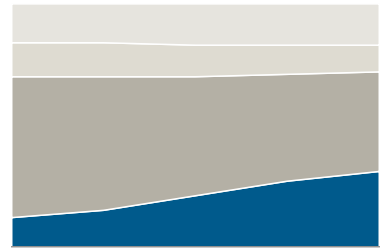
Volatile on top. A jumpy layer at the bottom makes every layer above it jump. Keep steady layers low.

Few layers. Four or five at most; fold the rest into "Other".

Overlapping areas

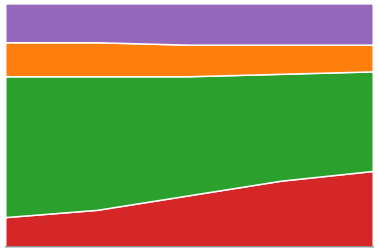


Try to avoid
Translucent layers over each other.

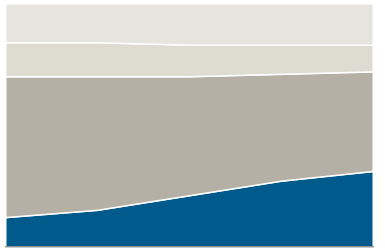


Recommended
Stacked layers.

Color order



Try to avoid
Unrelated bright colors.



Recommended
Blue story, ordered greys.

High variability

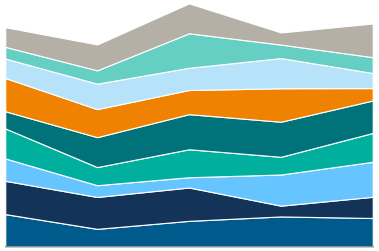


Try to avoid
The volatile layer at the bottom.



Recommended
Steady layers low, volatile on top.

Number of layers



Try to avoid
Nine thin layers.



Recommended
Three: story, comparison, rest.

CHANGE OVER TIME

Bubble timeline

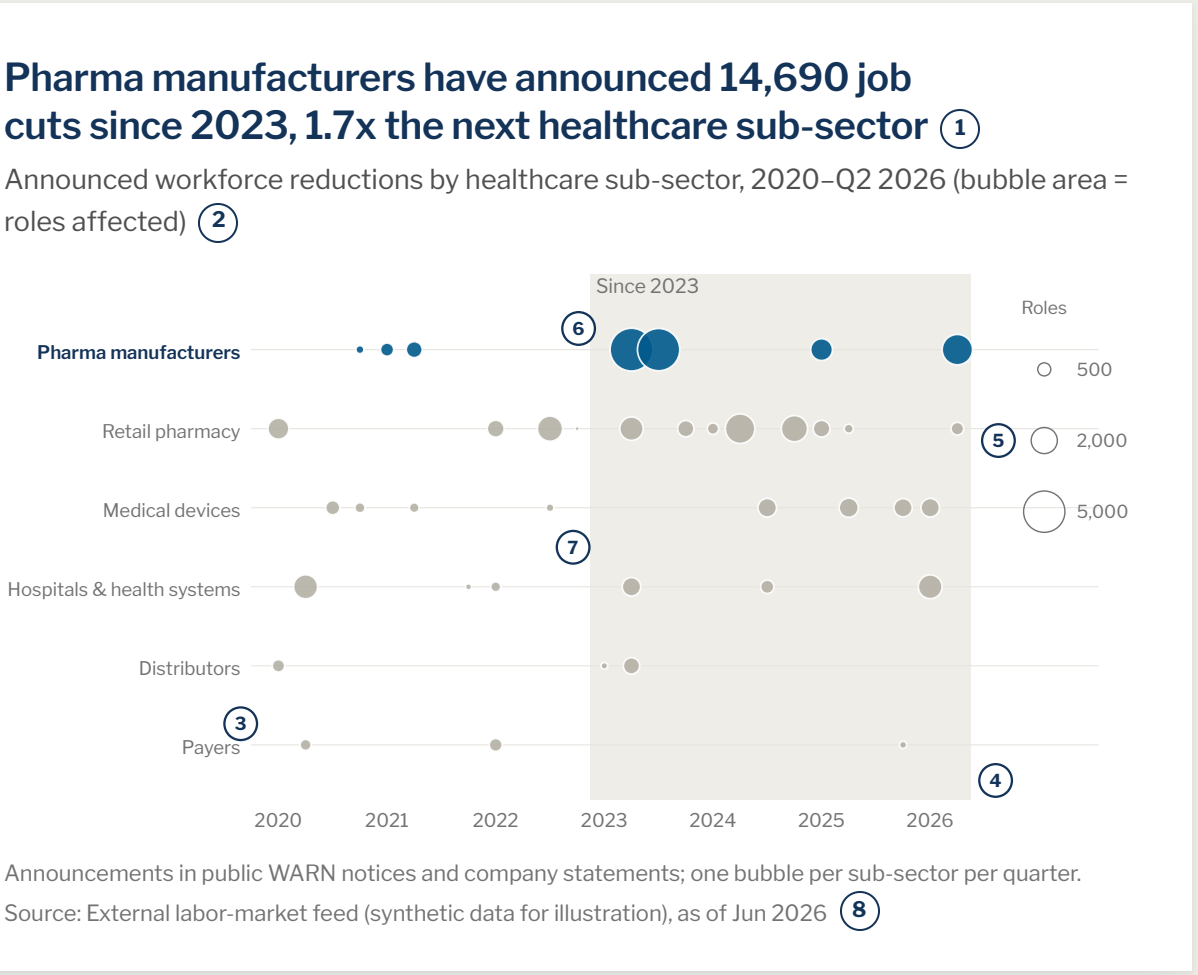
Events placed on a time axis, with bubble area showing size. It combines a timeline with a proportional-area chart.

Use it for discrete events of different sizes: announced layoffs in the talent market, reorganizations, hiring waves. Each category gets its own row.

McKesson spec

- Bubble area, not diameter, proportional to value
- One grey for all rows; the headline's row in blue
- Largest bubbles drawn first, with white outlines
- A size key with two or three reference circles

1. **Insight headline**
   Who stands out, by how much.
2. **Subtitle**
   Says what bubble area means.
3. **One row per category**
   Sorted by total, largest on top.
4. **Time along the bottom**
   Even intervals, years labelled.
5. **Size key**
   Reference circles; area ∝ value.
6. **Focus row**
   Blue; white outlines keep overlaps readable.
7. **The period that matters**
   Shaded and named.
8. **Footnote and source**
   What counts as an event.



CHANGE OVER TIME

Bubble timeline best practice

The rows already separate the categories; let size do the rest.

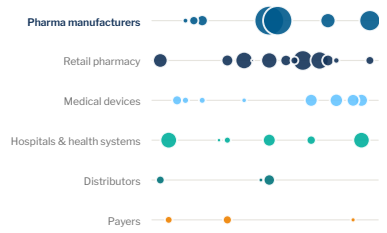
One color. Coloring each row repeats what the rows already say. Use grey, and blue for the headline's row.

Don't hide small bubbles. Draw the largest first, with white outlines.

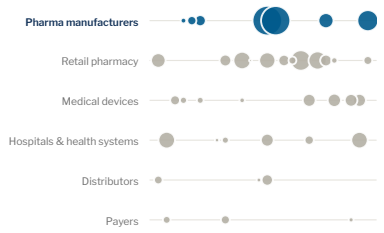
Sort the rows. By total, or by a meaningful order, not alphabetically.

Show the scale. Without a size key, readers can only say "bigger".

Coloring bubbles

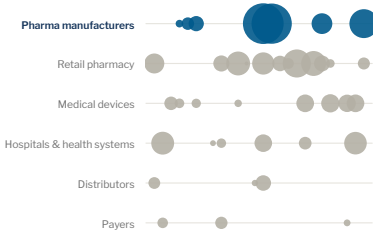


Try to avoid
A color per row.

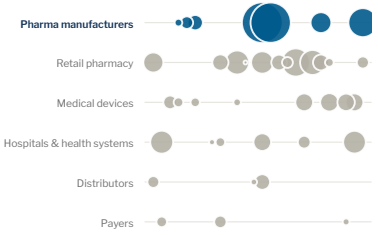


Recommended
Grey, one row blue.

Overlapping bubbles

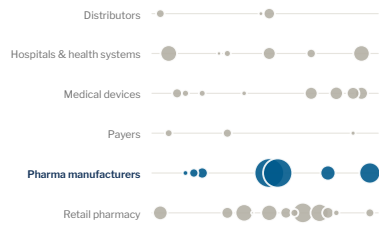


Try to avoid
Big bubbles on top.

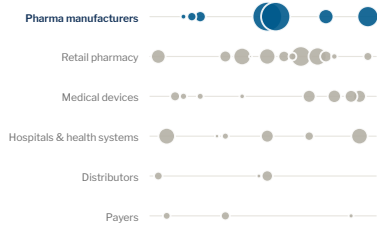


Recommended
Big behind, white outlines.

Row order

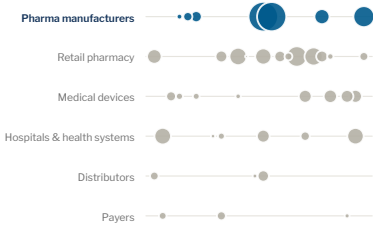


Try to avoid
Alphabetical rows.

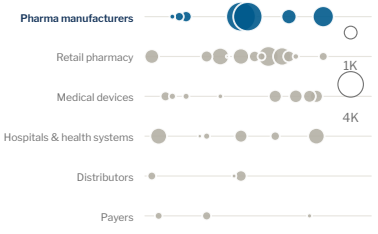


Recommended
Sorted by total.

Size key



Try to avoid
No way to read size.



Recommended
Two reference circles.

CHANGE OVER TIME

Small multiples

The same small chart repeated for each group, on one shared scale. The eye compares shapes across panels instead of untangling crossing lines.

Use it when six or more groups each need their own trend: business units, regions, job families. In each panel, the panel's own line is blue and the others sit faintly behind it.

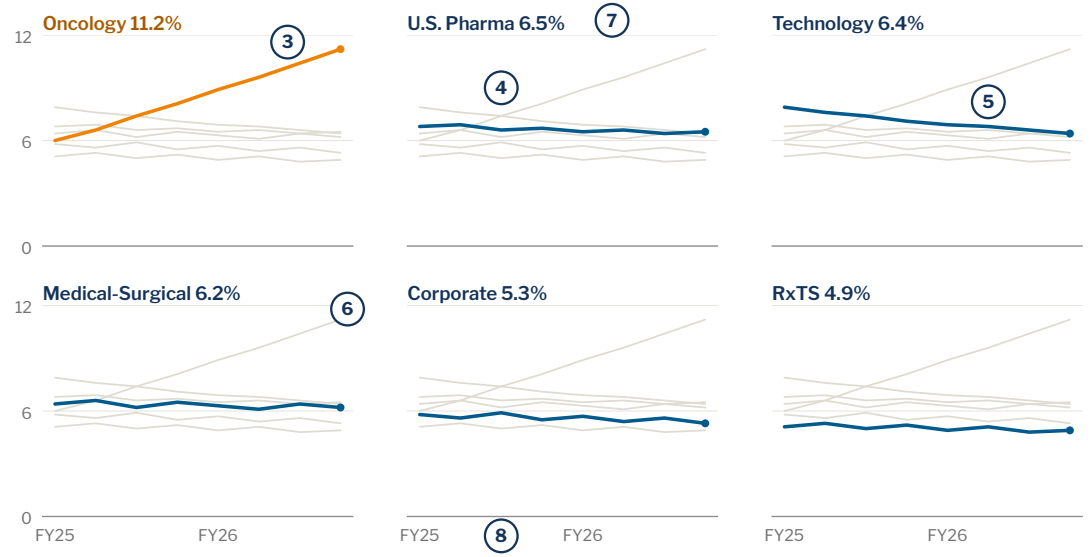
McKesson spec

- One shared scale, stated once
- Panels ordered by value or a natural order
- Panel title = name + latest value
- Axis labels only on the outer edge

1. **Insight headline**
   The one panel that differs.
2. **Subtitle**
   Metric, unit, period, once for all panels.
3. **The exception**
   In orange, because it is getting worse.
4. **Its own line in blue**
   The other units faint behind it, for context.
5. **Ordered by value**
   Highest latest value first.
6. **Shared scale**
   0–12% in every panel.
7. **Title carries the number**
   Name and latest value.
8. **Outer axes only**
   Labels on the left column and bottom row.
9. **Footnote and source**
   Define the metric.

Oncology regretted attrition nearly doubled in two years while every other business unit stayed within 2 points (1)

Regretted attrition by business unit, Q1 FY25–Q4 FY26 (% of high performers, annualized) (2)



Regretted attrition = voluntary exits of employees rated "exceeds" in the last review, annualized.

Source: Workday HCM (synthetic data for illustration), as of 31 Mar 2026 (9)

CHANGE OVER TIME

Small multiples best practice

Small multiples only work if every panel is read the same way.

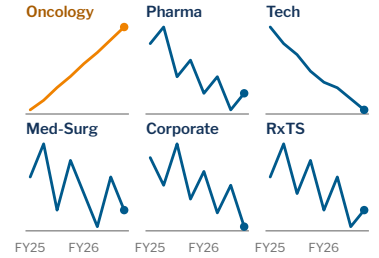
One scale. Independent scales make a 0.5-point wobble look like the 5-point rise.

Context in every panel. The other units, faint behind each line, show whether a panel is unusual.

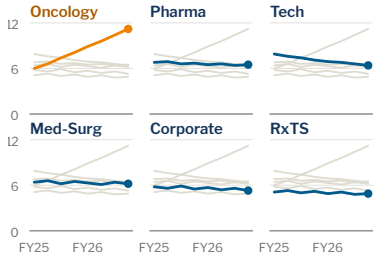
Order with meaning. By value, size or geography, not the alphabet.

Label the edges only. Repeated axes are clutter the eye reads six times.

Shared scale

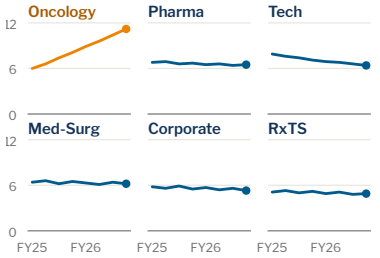


Try to avoid
Every panel zooms itself.

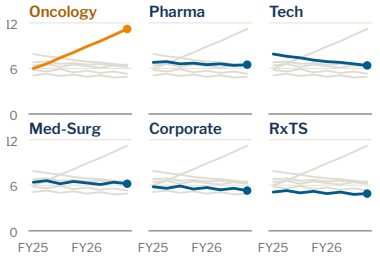


Recommended
One scale; the rise stands out.

Context

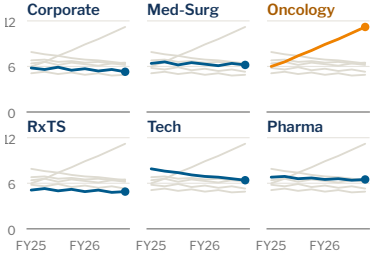


Try to avoid
Lines alone.

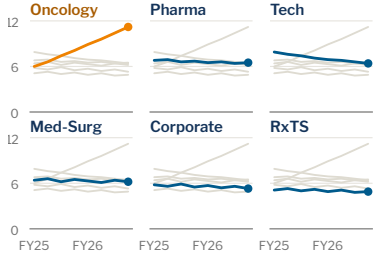


Recommended
The others faint behind.

Panel order

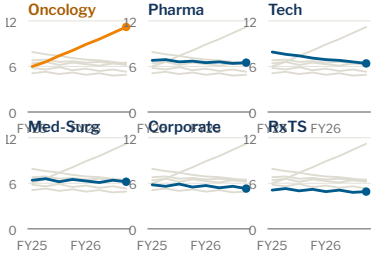


Try to avoid
Alphabetical.

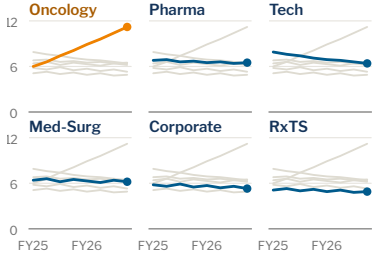


Recommended
By latest value.

Axes



Try to avoid
Axes on every panel.



Recommended
Outer edge only.

CHANGE OVER TIME

Build change-over-time charts

Native, editable charts in every tool. Tested code and click-by-click steps are on the cookbook pages.

line.md · stacked-area.md · bubble-timeline.md · small-multiples.md

Before you send it

- [ ] Time left to right, at true intervals
- [ ] One line or layer in blue, matching the headline
- [ ] Names at line ends; no legend
- [ ] Forecasts dashed, labelled and with a range
- [ ] Small multiples share one scale

PowerPoint and Excel	**Line:** Insert › Line; format the focus series 2.25 pt Focus blue, the rest 1.25 pt grey; delete the legend and add data labels to the last point only (Series Name + Value). Use a *date* axis so missing periods keep their space. **Stacked area:** Insert › 100% Stacked Area; story series first in Select Data; white 1 pt borders. **Small multiples:** build one chart, fix the axis bounds (Minimum/Maximum, not Auto), then copy it per group.
Power BI	Line chart with the focus series colored by measure; turn off the legend and use *Series labels*. Forecast bands as low/high measures shown with *Error bars* › *Shaded area*. Small multiples are built in: add the group to *Small multiples* and keep *Y-axis* › *Shared*.
Python	matplotlib with `mckviz`; `mv.label_line_ends(ax)` places and nudges end labels; `mv.shade_period(ax, start, end, label)` marks the period.

```
for fam, s in df.items():                # one column per job family, one row per quarter
    ax.plot(x, s, label=fam, color=C.focus if fam == focus else C.context,
            lw=2.4 if fam == focus else 1.4)
mv.shade_period(ax, 6, 11, "After the reset")
mv.label_line_ends(ax, fmt="{:.1f}")      # names at the ends, nudged apart
```

HTML	Plotly `scatter` traces with `mode: "lines"`, an annotation per line end, and a `shape` for the shaded period; `facet_col` (Plotly Express) for small multiples with shared axes.
R	`geom_line()` with `mck_highlight()` colors and `geom_text()` end labels; `geom_area()` for stacks; `facet_wrap(~ unit)` keeps a shared scale by default.